

AI in Planning and Project Development



Presented by: Grace Sammartino
Senior Account Executive, Transportation Division, Esri
August 2026

AI in Planning and Project Development

Agenda

- Company Overview
- AI Platform and Framework
- Showcase
 - GeoAI Tools and Examples
 - AI Assistant Examples
 - Agentic AI



A world map with a dark grey background. Numerous small dots in orange and yellow are scattered across the landmasses, representing the global distribution of Esri users. The dots are particularly dense in North America, Europe, and East Asia.

esri

World Leader in Geographic Information Systems (GIS)

Over 10 Million Users

33,000 Businesses

12,000 Colleges & Universities

8,600 Utilities

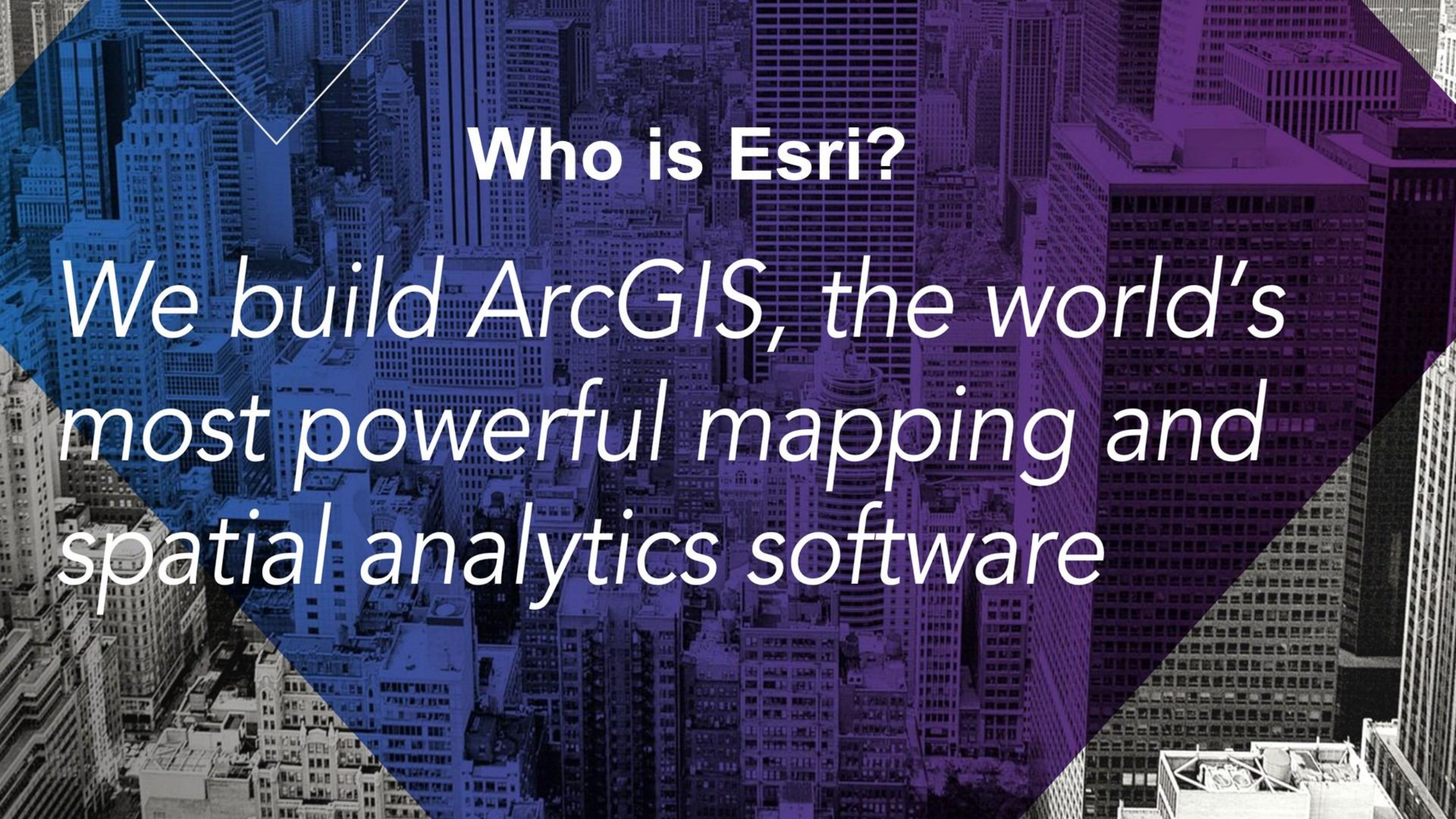
12,600 NGOs

1,200 National Government Agencies

12,000 States & Regional Agencies

30,800 Cities & Local Governments

Empowering Over 680K Organizations Worldwide

An aerial photograph of a dense city skyline, likely New York City, featuring numerous skyscrapers. The image is overlaid with a large, semi-transparent geometric shape that is blue on the left and purple on the right, separated by a diagonal line. A white line also extends from the top left corner towards the center.

Who is Esri?

*We build ArcGIS, the world's
most powerful mapping and
spatial analytics software*

ArcGIS

Data Fusion Platform

Dashboards

Performance
Management

IoT

Autonomous
Vehicles

Real-Time

Analytics

Situational
Awareness

Sensors

Artificial
Intelligence

Environmental
Monitoring

Machine-Learning



What Is a GIS?

A **System** for Managing, Sharing and Applying Geographic Information



ArcGIS is a System of Systems

Connecting Department of Transportations for Common Goals



Action

Understanding

Asset Management

Capital Projects

Traffic Operations

Safety

Maintenance Operations

Incident Management

Planning & Performance

Environmental

Multimodal

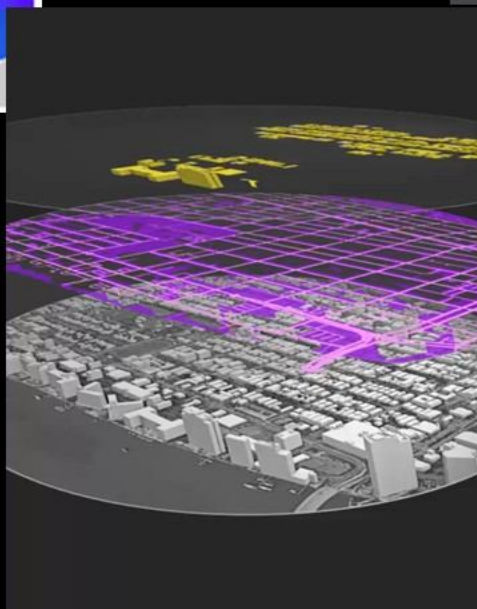
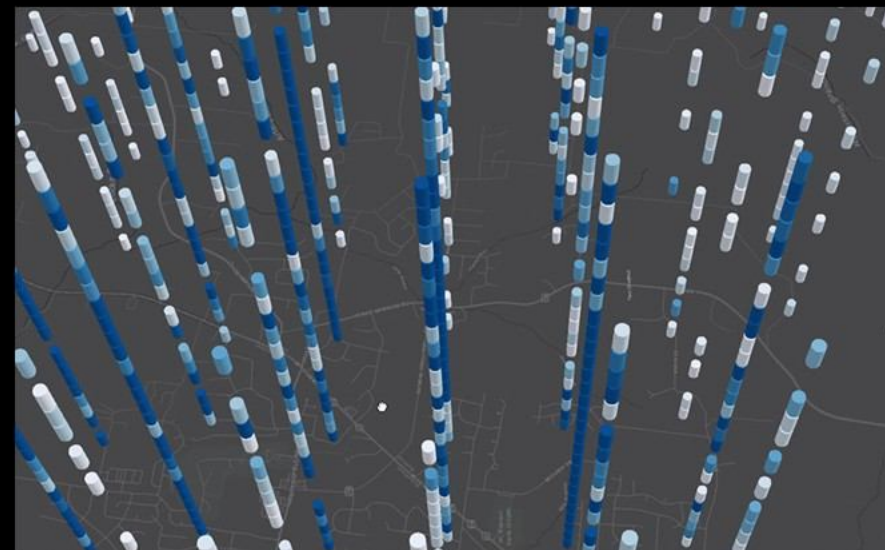
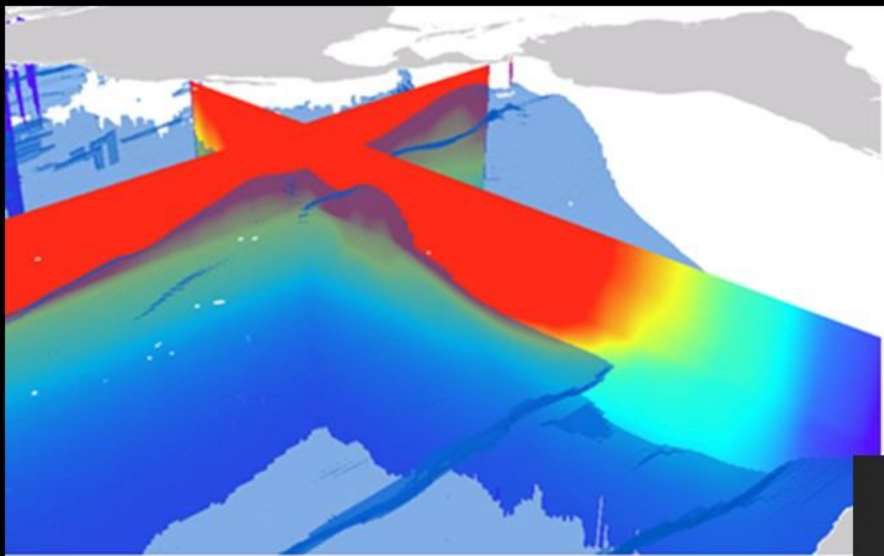
Digital Delivery

Equity & Access

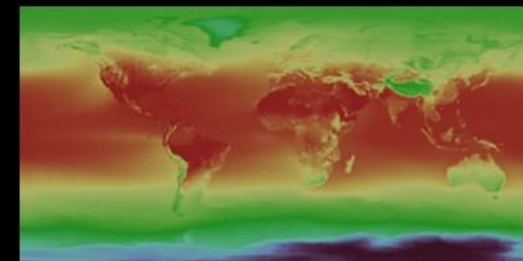
Climate Resilience

Predictive Analytics

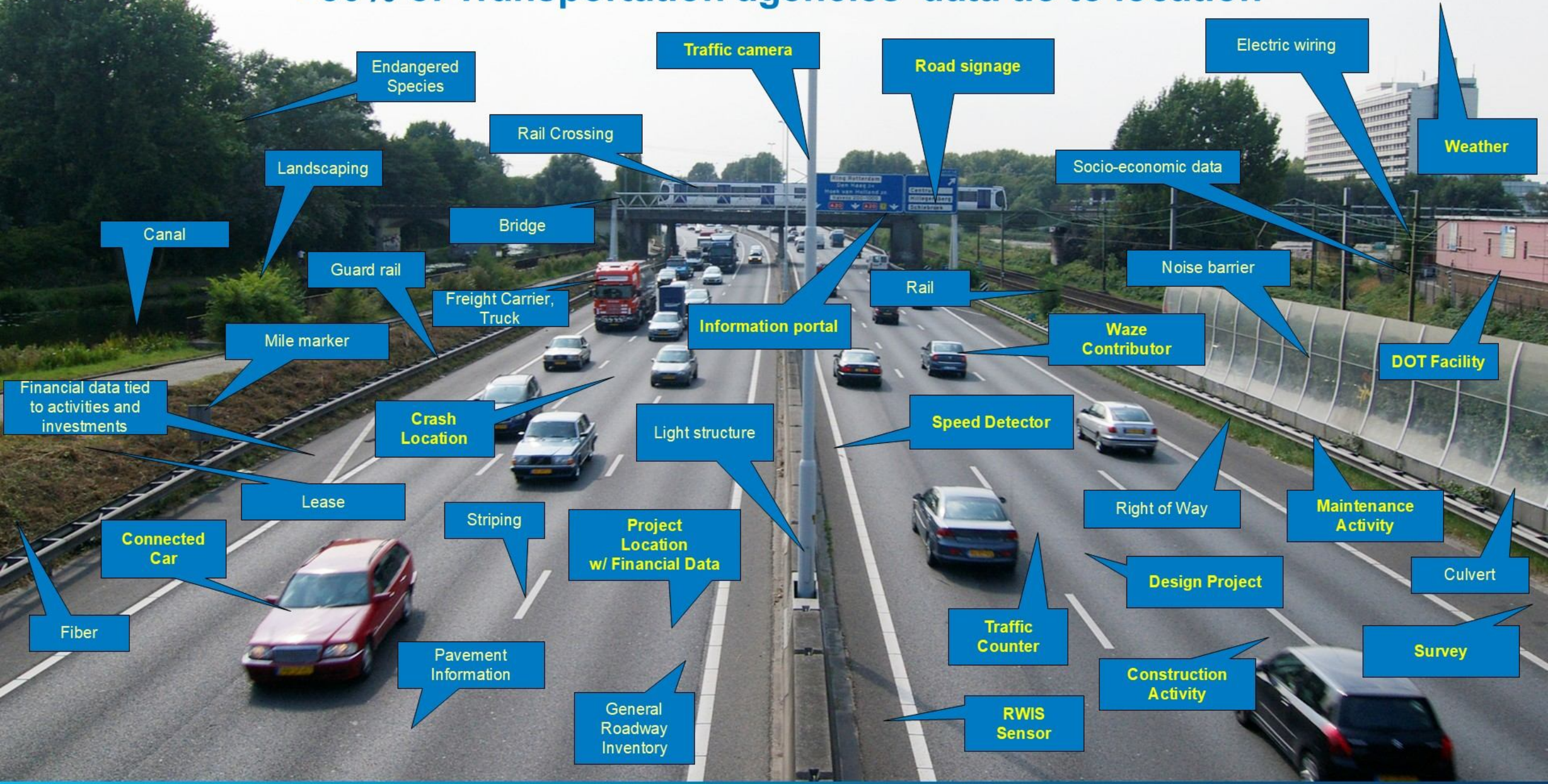
...Enabled by AI-ready enterprise GIS and integrated data.



ArcGIS Platform
Ability to transform data into
actionable information



Estimates are that >80% of Transportation agencies' data tie to location



Estimates are that >80% of Transportation agencies' data tie to location

Spatial is Special



AI is used today literally everywhere..

Autonomous Cars



Sentiment Analysis



Chatbots



War Robots



Advanced Video Analytics



Crime Prediction



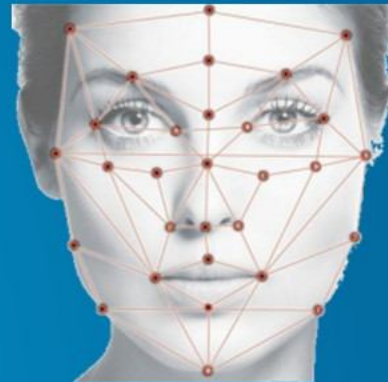
Predictive Maintenance



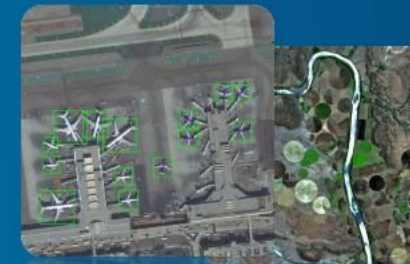
Cancer Detection



Facial Recognition



Advanced Satellite Intelligence



Stock Market Prediction



Personalized Marketing



ArcGIS Is Advancing

Evolving Capabilities, Improving Quality . . .
and Meeting Emerging Needs



AI is integrating and evolving with many other innovations

JUN 2008

Geospatial ML

MAR 2017

Geospatial DL R&D

JUL 2018

Geospatial DL for Analysis
& Data Extraction

JAN 2019

DL in Python API, (*learn module*)

OCT 2020

Pre-Trained AI Models

NOV 2020

DL on multispectral imagery

NOV 2020

DL used in 3D Basemaps Solution

April 2021

DL Object Tracking for FMV

JUL 2021

DL comes to ArcGIS Online

2008 - 2021



JUL 2022

Deep Learning Studio

OCT 2022

GeoAI Toolbox in Pro, AutoML, Text Analysis

FEB 2023

AutoDL and Multi-feature extraction

2022 - 2023

2023 - now

Today

Agentic AI

H1 2024

Generative AI
Assistants in ArcGIS

FEB 2023

Smart Assistants
in Survey123

NOV 2023

Multi-modal AI, Bias
Mitigation in AutoML

JAN 2024

SAM and Prithvi – foundational models
for Geospatial AI

MAR 2024

Text SAM – Vision Language
foundational models

MAY 2024

LLM support for Text AI in arcgis.learn

AI is not one product. It spans ArcGIS.



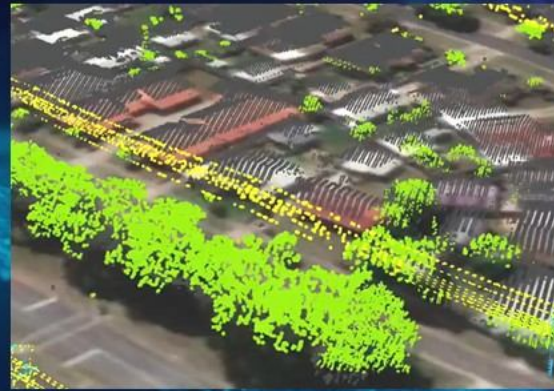
What Value Does AI Bring to ArcGIS?

GENERATE EFFICIENCIES



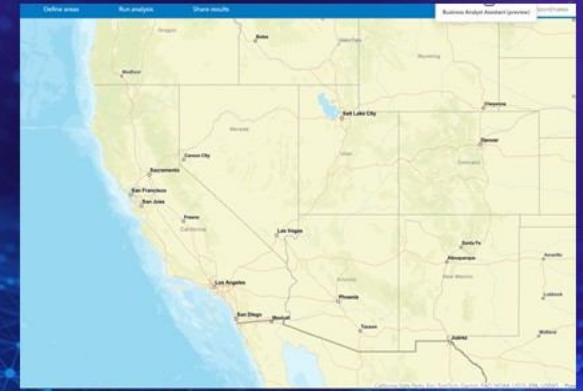
Connect Business Systems
Democratize Data Creation
Automate Manual Tasks

ACCELERATE ANALYSIS



Deeper Insights
Visualization tools
Predictive Spatial Analytics

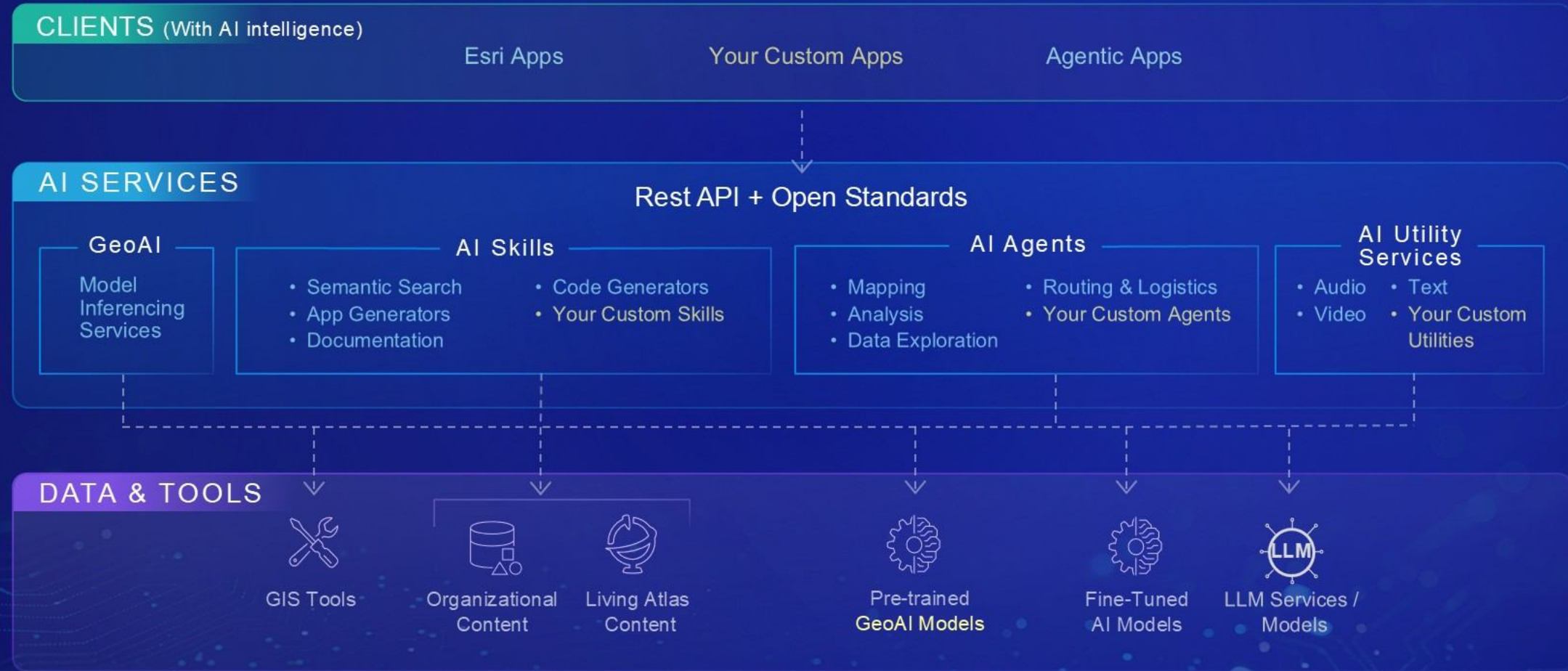
ENHANCE USER EXPERIENCE

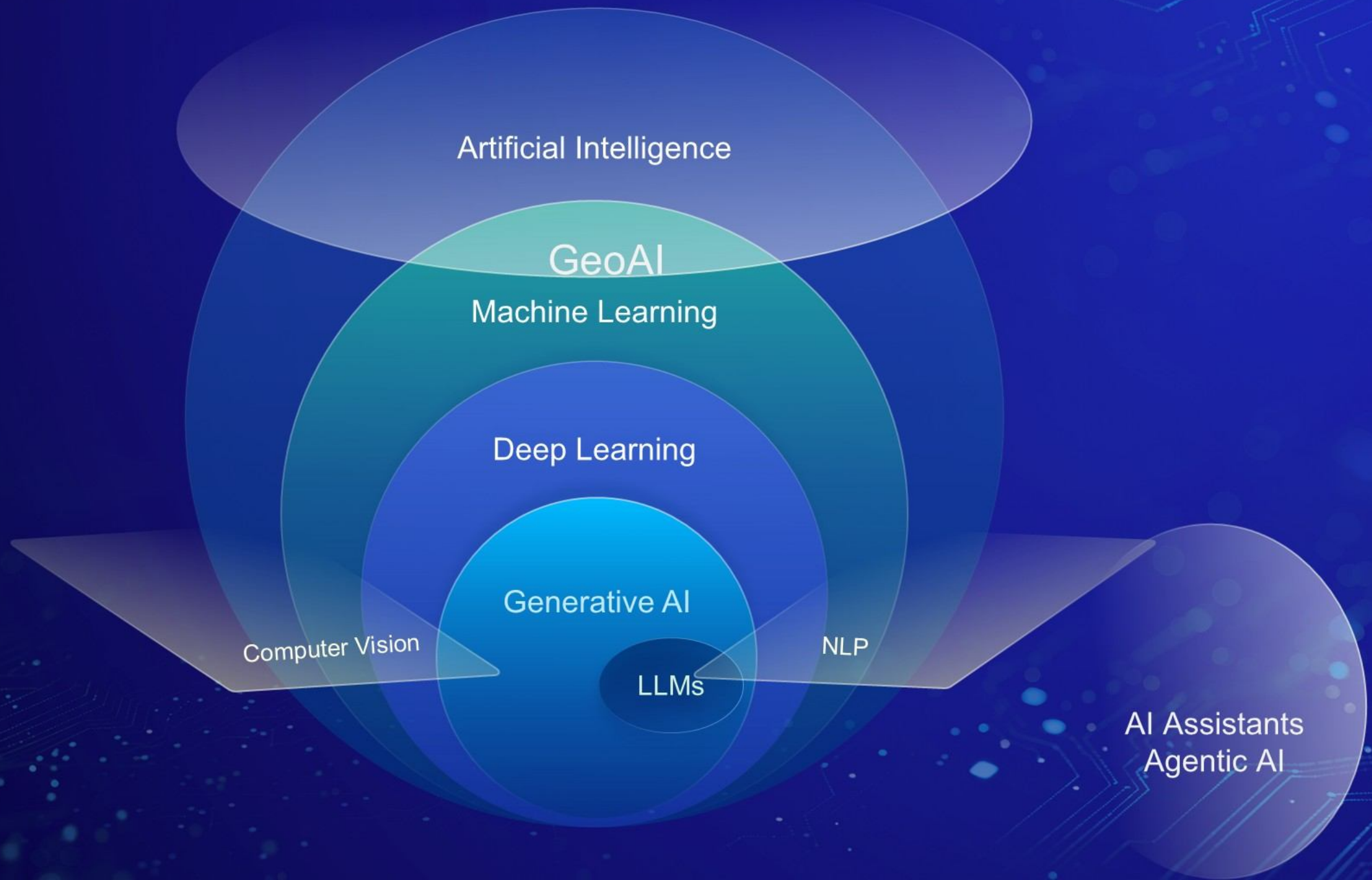


Intelligent Experiences
AI-enhanced mobile and office tools
Virtual and In-App Assistants



ArcGIS as an **Extensible** Geospatial AI Platform





ArcGIS Is Being Enhanced with AI

Integrating the power of GIS and AI across organizations and workflows



AI Tools & Models

*Advancing the
Science of GIS*

Science



AI Assistants

*Enhancing user
experiences*

Productivity



Agentic AI

*Automating tasks &
expanding GIS*

Impact



ArcGIS

- GIS Tools & Data
- Agents
- Models
- LLMs

ArcGIS Is Being Enhanced with AI

Integrating the power of GIS and AI across organizations and workflows



AI Tools & Models

*Advancing the
Science of GIS*

Science



AI Assistants

*Enhancing user
experiences*

Productivity



Agentic AI

*Automating tasks &
expanding GIS*

Impact



ArcGIS

- GIS Tools & Data
- Agents
- Models
- LLMs

Making AI Approachable

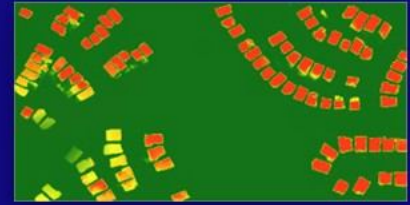
Ready-to-use Pretrained AI models

- Cars
- License Plate Blurring
- Parking Lots
- **Land Cover**
- Parking Spots
- Humans
- Crowd Counting
- Face Blurring
- **Buildings**
- Roads
- Parcels
- Ag Fields
- Swimming Pools
- Well Pads
- Oil Spills
- Palm Trees
- Power Lines
- **Transmission Towers**
- Insulator Defects
- Wind Turbines
- Solar Arrays
- Solar Panels
- Ships
- Shipwrecks
- **Segment Anything Model (SAM)**
- Trees
- Plant Leaf Disease
- Common Object Detection
- Text Parsing from Photo
- **Object Tracking**

new

- Canopy Height Estimation
- **Hugging Face Hub Integration**
- Depth Anything
- **Prompt-Based Segmentation**
- Pedestrian Infrastructure Classification
- Wildfire and Smoke Classification
- Wildfire Delineation
- **Damage Classification**
- **TextSAM**
- CLIP Zero-Shot Classifier
- GroundingDINO
- Damage Assessment (Drones)
- **Map Simplification**
- Edge Detection
- **Vision-Language Context-Based Classification**
- Building Point Classification
- Building Change Detection

... Automating Workflows
... Creating New Data Layers

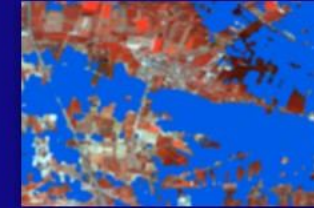


Cars (SAM)



Change Detection
Buildings

Flood Segmentation



Clouds



Building
Footprints



Crop Classification (Prithvi)



Map Simplification



Land Cover

ArcGIS is a Comprehensive GeoAI Platform

All types
of data



Capabilities

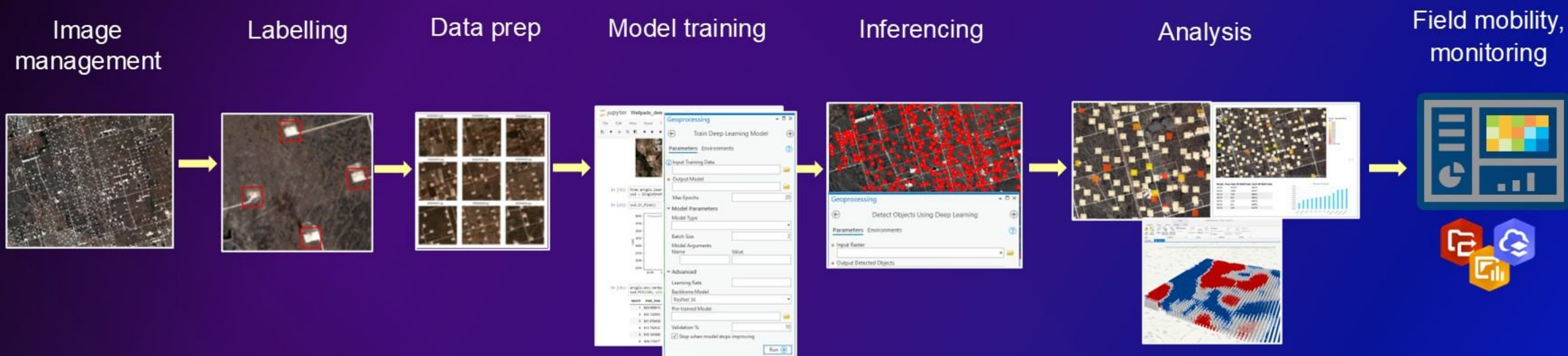


Take action

... Supporting all aspects of AI

Imagery AI: End-to-End Workflow

Extract Insights from Imagery at Scale, with High Speed and Accuracy



For Wide Range of Data Types

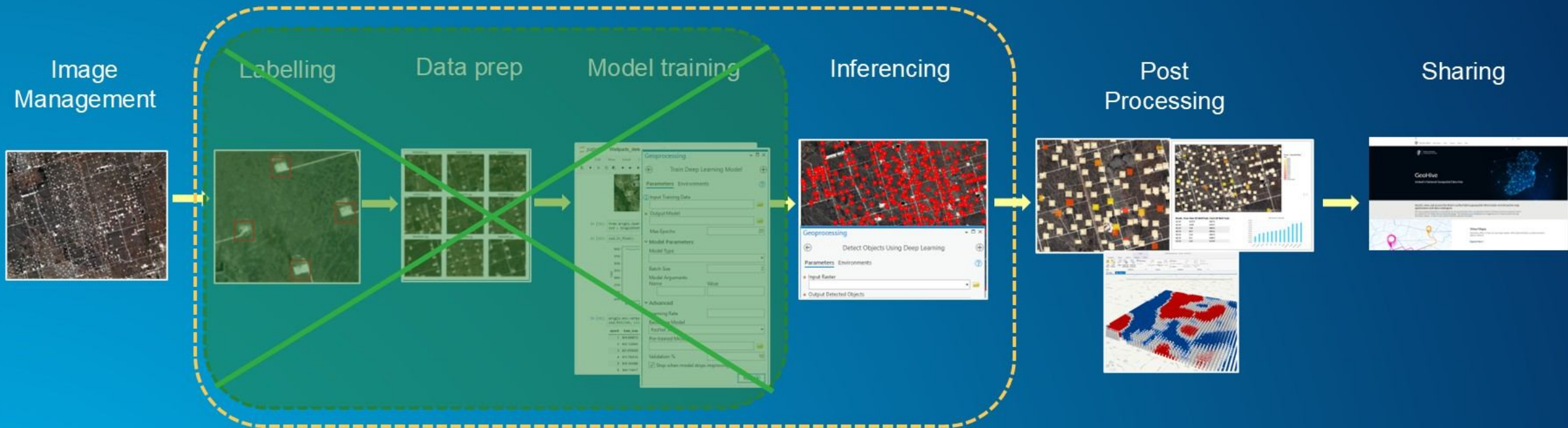
- Aerial
- Satellite
- Radar
- Lidar
- Motion imagery
- Bathymetry
- Point cloud
- Drone

Implementing Many Tasks

- Object classification
- Object detection
- Pixel classification
- Image translation
- Object tracking
- Scanned maps

Deep Learning Workflow

Integrating AI & GIS



For Wide Range of Data Types

- Aerial
- Satellite
- Radar
- Lidar
- Motion imagery
- Bathymetry
- Point cloud
- Drone

Implementing Many Tasks

- Object classification
- Object detection
- Pixel classification
- Image translation
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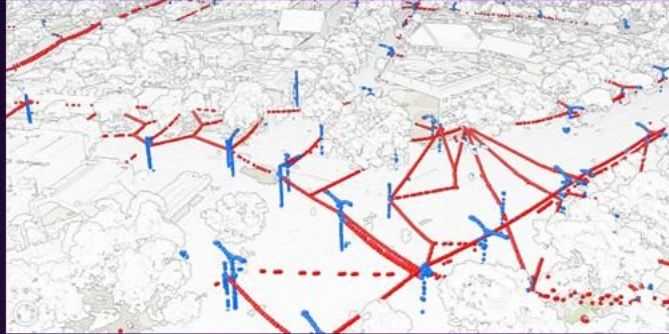
Integration with DL Libraries



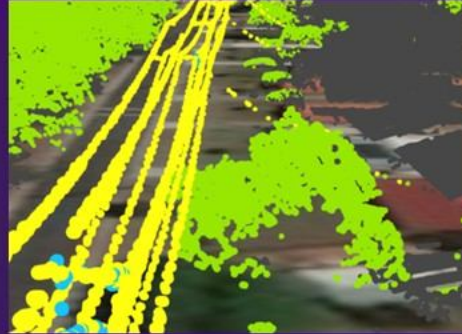
Feature Extraction from Lidar

Data Pre-processing, Labelling, Training, and Inference

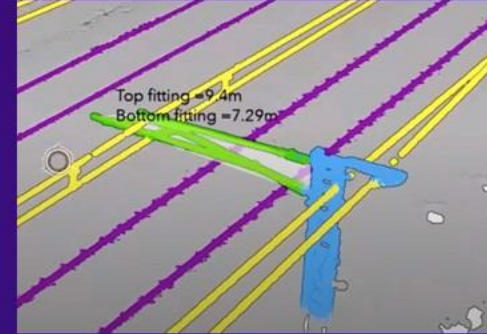
Utility poles and lines



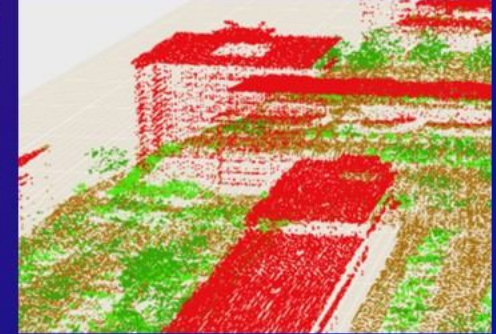
Encroachments and trees



Rail assets



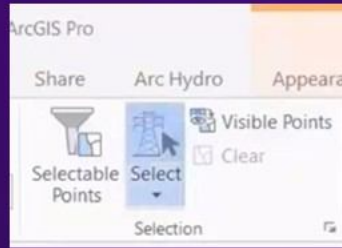
Buildings



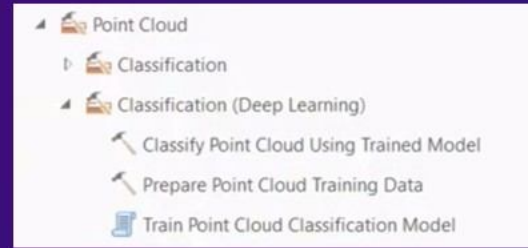
Streetlights



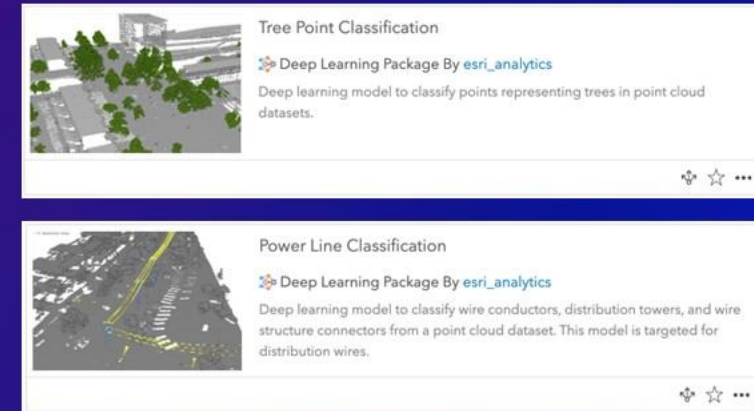
Labelling tools



Training and inference tools



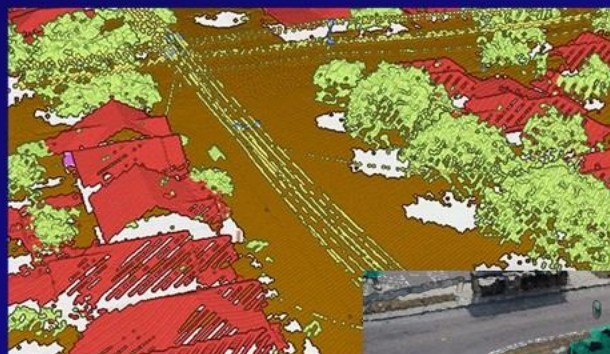
Pre-trained models (ArcGIS Living Atlas)



GeoAI: 3D Deep Learning

3D Deep Learning Tasks

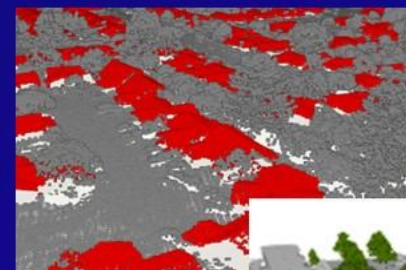
- Point Cloud Classification
 - PointCNN
 - RandLANet
 - SQNSeg
- 3D Object Detection
 - MMDetection3D



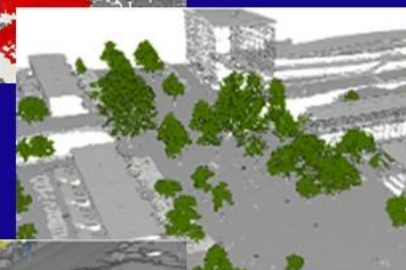
Point Cloud Classification



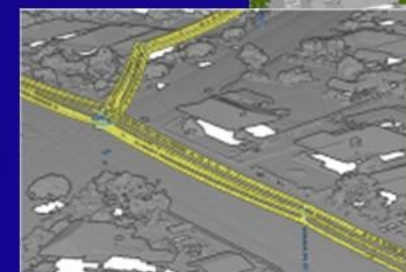
3D Object Detection



Building Point Classification



Tree Point Classification



Power Line Classification

Pretrained Models

- Building Point Classification
- Tree Point Classification
- Power Line Classification

Pretrained AI Models

5. Extract Insights from Unstructured Text

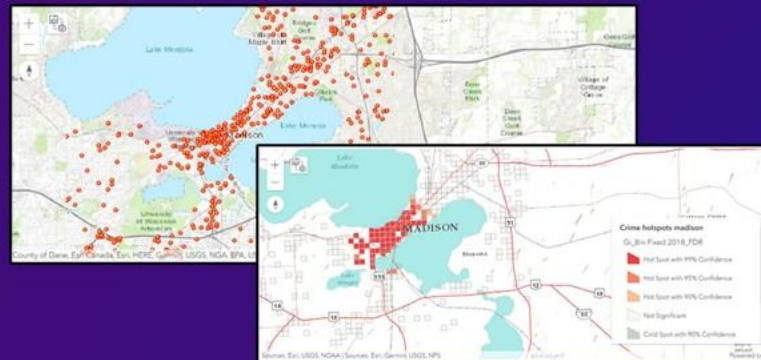
Extract Entities, Classify Text, Translate StoryMaps, and More

Extracting entities and relationships from text reports

Address a Crime c Crime_datetime C-c Reporting_officer C-r Weapon M

Reported_date d Reported_time t

A Radcliffe Drive resident said three men - at least two of whom were carrying handguns - entered her apartment early this morning. They ordered her to give them the combination to a safe which they took from her home. She was not injured. Detectives believe this was a targeted armed robbery. Released 08/07/2018 at 11:17 AM by PIO Joel Despain



Classifying country of incomplete addresses

Address	Country
40, Rue Eugene Delacroix, Seloncourt, 25230	FR
736, E AMOROSO PL, 90291	US
15127, CALLE EDGAR ALÁN POE, CHIHUAHUA, Chihuahua, Chihuahua	MX
25376, COSTEAU ST, LAGUNA HILLS, CA	US
1-6, 一乗寺堀 / 内町	JP
3701-1, 飯香浦町	JP
30, CL CARMEN AMAYA, 39011	ES
NW21-17-23W	CA
8140, avenue Belges	CA
80, WURUNDJERI BOULEVARD, BERWICK, VIC, 3806	AU

Translate StoryMaps

Introduction High-Injury Roadway Alcance actualizado Preguntas

The Preliminary Engineering Report for a new bridge will advance into the design phase of project. The bridge will advance into the design phase of project. The bridge will advance into the design phase of project.

As part of the community conversation to develop improvements were also identified. A final set of improvements was identified. A final set of improvements was identified.

Healthy

Currently, a trail of travel se encuentra en South Pleasant Valley Road para caminar y andar en la parte de la ciudad de Austin. Calles activas durante la pandemia de COVID-19. La COVID-19 ha llevado al aumento del tráfico de senderos y trail traffic. El aislamiento social. Esta iniciativa no

Correct and normalize mistyped street addresses

MISTYPED NON-STANDARD ADDRESS	NORMALIZED ADDRESS
940, NORTH PENNSYLVANIA AVNEUE, MASON ICTY, IOWA 50401	940, N PENNSYLVANIA AVE, MASON CITY, IA 50401
220, SOYTH RHODEISLAND AVENUE, MASON CITY, IOWA 50401	220, S RHODE ISLAND AVE, MASON CITY, IA 50401

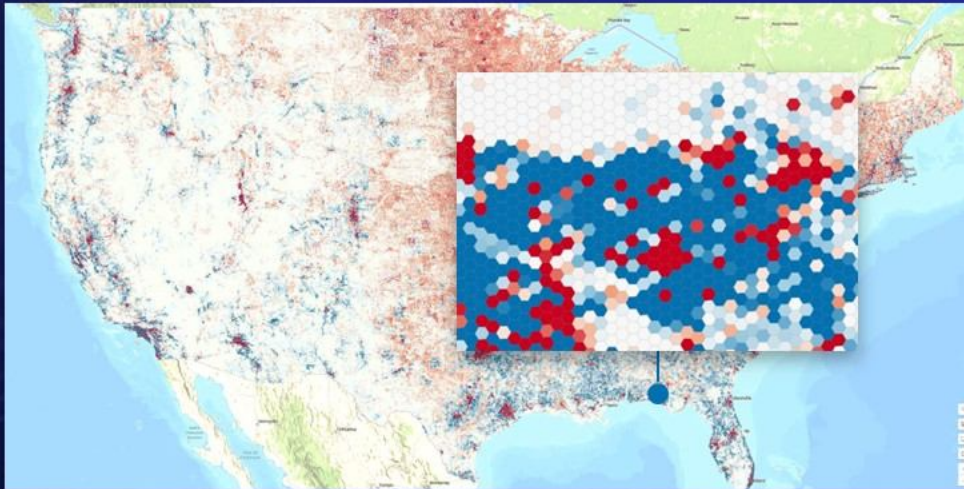
Location Embeddings

Augmenting spatial analysis with additional contextual information

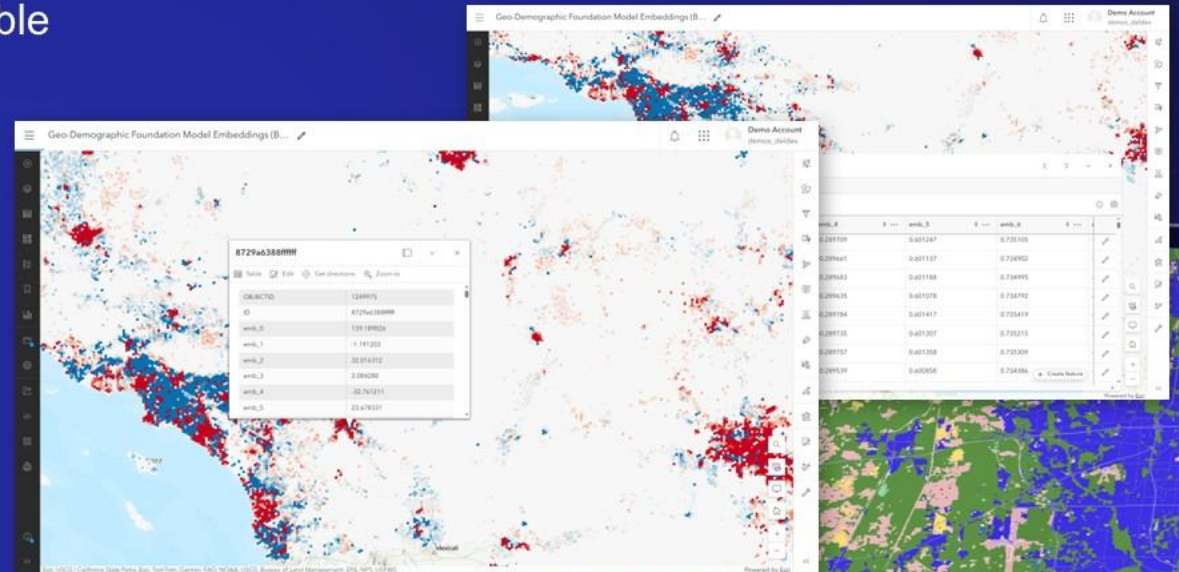
Location embeddings are machine-understandable representations of **GEOGRAPHY**

... Providing deeper spatial insight
and better predictions

Geo-Demographic Location Embeddings



Numerical representations of relationships and patterns in geographic data



Supporting ...

Similarity Search
Pattern Recognition
Change Detection
Clustering

Deep Learning Studio

Bring the entire deep learning workflow into a focused environment

Coming

- Run in ArcGIS Online
- Kubernetes deployment
- Integrated spectral profile
- Support time-series
- Model assisted labelling
- New metadata formats

ArcGIS Enterprise +
ArcGIS Image Server



Project-based Workflows



Prepare
Training
Data



Train Model



Run
Inferencing

- Scalable multiuser workflows
- Small to large projects
- Project management

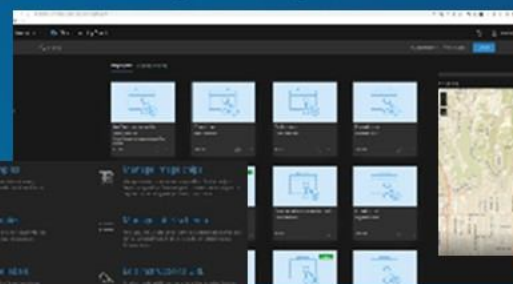
USDA
Fugro

Deep Learning Studio

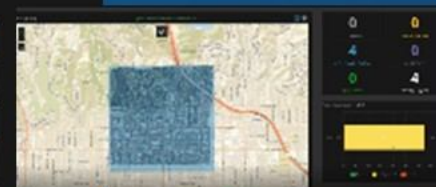
- Intuitive web application
- End-to-end deep learning pipeline
- Collaborative training data collection

Simplify and streamline AI Workflows

Project Management



Status Dashboard



Collaborative Experience



What **Workflows** Can These Tools Enable?





Prediction	Pattern Detection
	
Predicting Geospatial Events/Phenomena	Finding Statistically Significant Clusters & Patterns

Accidents Prediction

Predict Accident Probability per Segment per Hour

What would Cause an Accident?



Temperature
Sun, Mon, Fri..



Wind Speed
Fast, Slow..



Visibility
High/Low



Snow Depth
High/Low



Day of the Week
Sun, Mon, Fri..



Time of the Day
12:45, 23:00



Month
Feb, Dec..



Road Width
20-30 M



Road Alignment
Straight / Curved



**Proximity to
Intersections**



Speed Limit
120 km/h



Sun Direction
East, West



Daily Traffic
AADT



**Proximity to
Billboards**

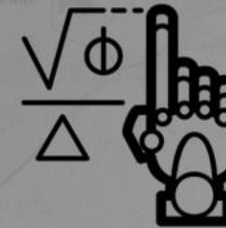
...

**10s of
Variables**

7 Years of Data
400,000 Accidents
500,000 Segments



**Impossible
to Manually
Analyze**

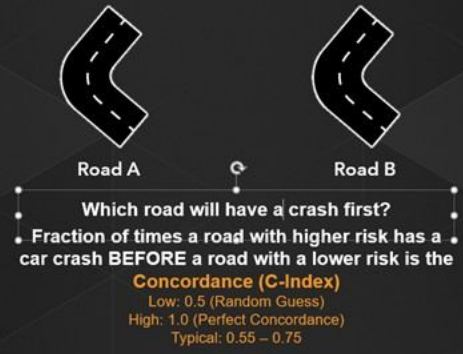


**Train a
Machine to
do?**

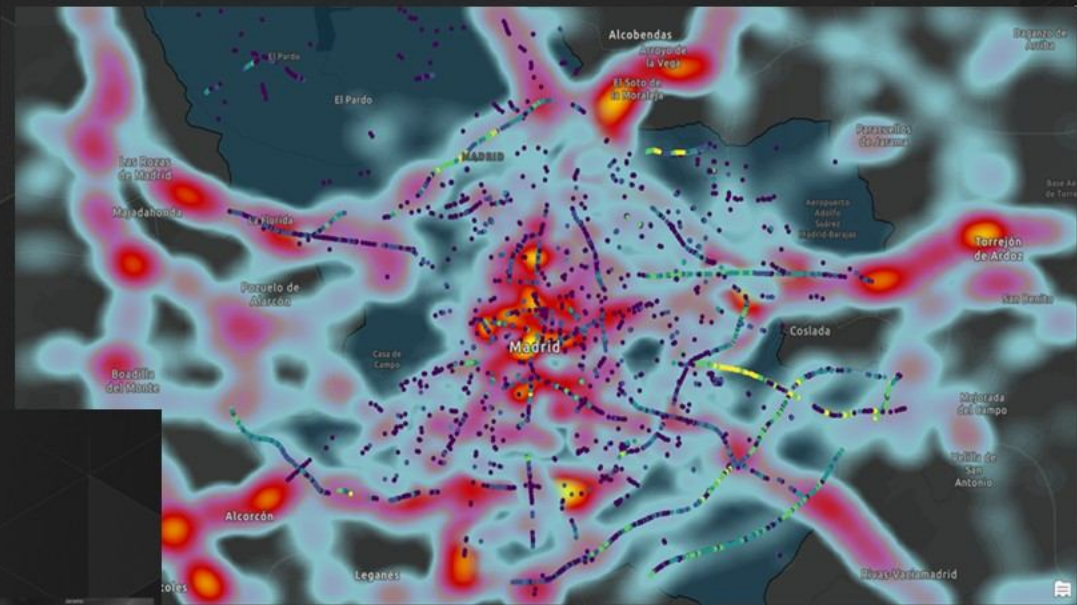


Blackspot Analysis POC for Predictive Analytics on Crash Risk

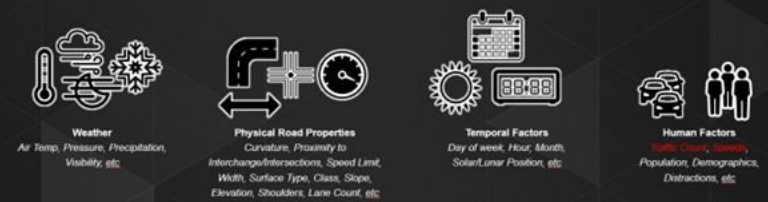
Model Evaluation



Terrain Data



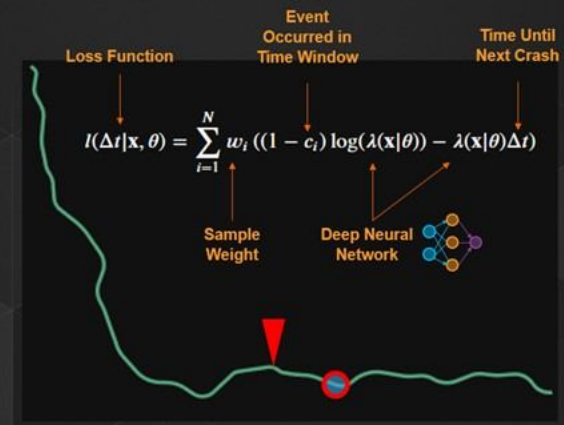
What could cause a car crash?



Research shows that dozens factors contribute, but the relationship is nonlinear.

Our (and others') past experience shows there is some predictability, but for the most part accidents are a random event

Our approach



Esri POC-For demonstration purposes only

Survey123 AI Assistant



Survey123



BETA

Sign Assessment Survey

Date/Time

Feb 24, 2025 at 1:36 PM

Take a photo of the sign

Drop image here or select image

Description

What is the classification code?

What are the colors?

What are the colors?

Brown and White

What is the shape?

Square

What is the category?

Guide/Information

Is it reflective?

Yes

What is the post type?

Metal

What is the post shape?

Round

Damage Assessment

survey123beta.arcgis.com

Create surveys using natural language

Hi, I'm Survey123 Assistant. Tell me your survey topic, idea, description, or any specific requirements. I can convert your ideas into a survey draft in designer page.

Imagine you work for the City of Redlands. Create a questionnaire to help citizens report damage to their houses after a recent storm event

Redlands Parks Incident Report - v2

Overview Design Collaborate Analyze Data Settings

English

Location

Find address or place

Lat: 34.044041 Lon: -117.178055

Comments

Text, number, date, and time

- Singleline text
- Multiline text
- Email
- Website
- Number
- Date
- Time
- Date and time

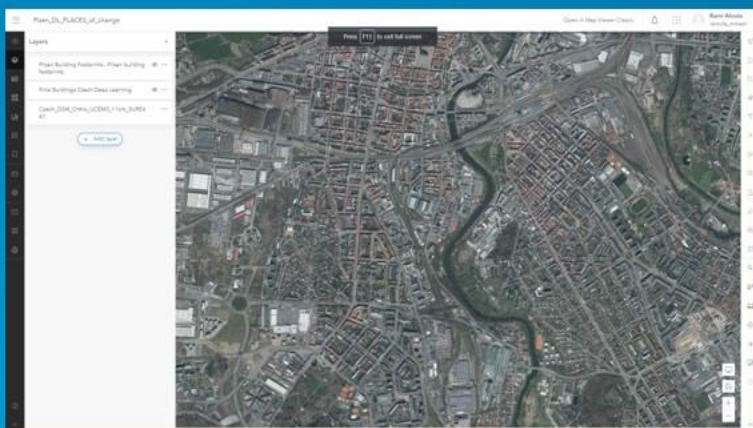
Choice

- ☒ Single choice
- ☐ Multiple choice
- ☒ Single choice grid
- ☐ Dropdown
- ☐ Likert scale
- ☐ Rating
- ☐ Ranking

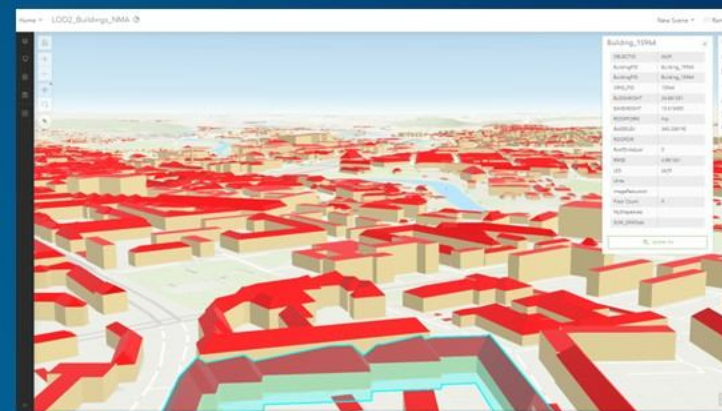
Save Preview Publish

Automating Digital Twins

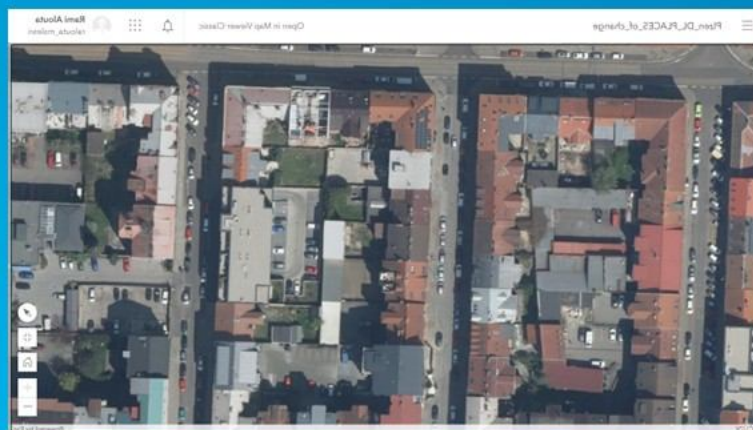
Czech National Mapping Agency (NMA)



Extracted Buildings using DL from True Ortho



Generate LOD2 Buildings



DSM; True Orthos; 3D Mesh



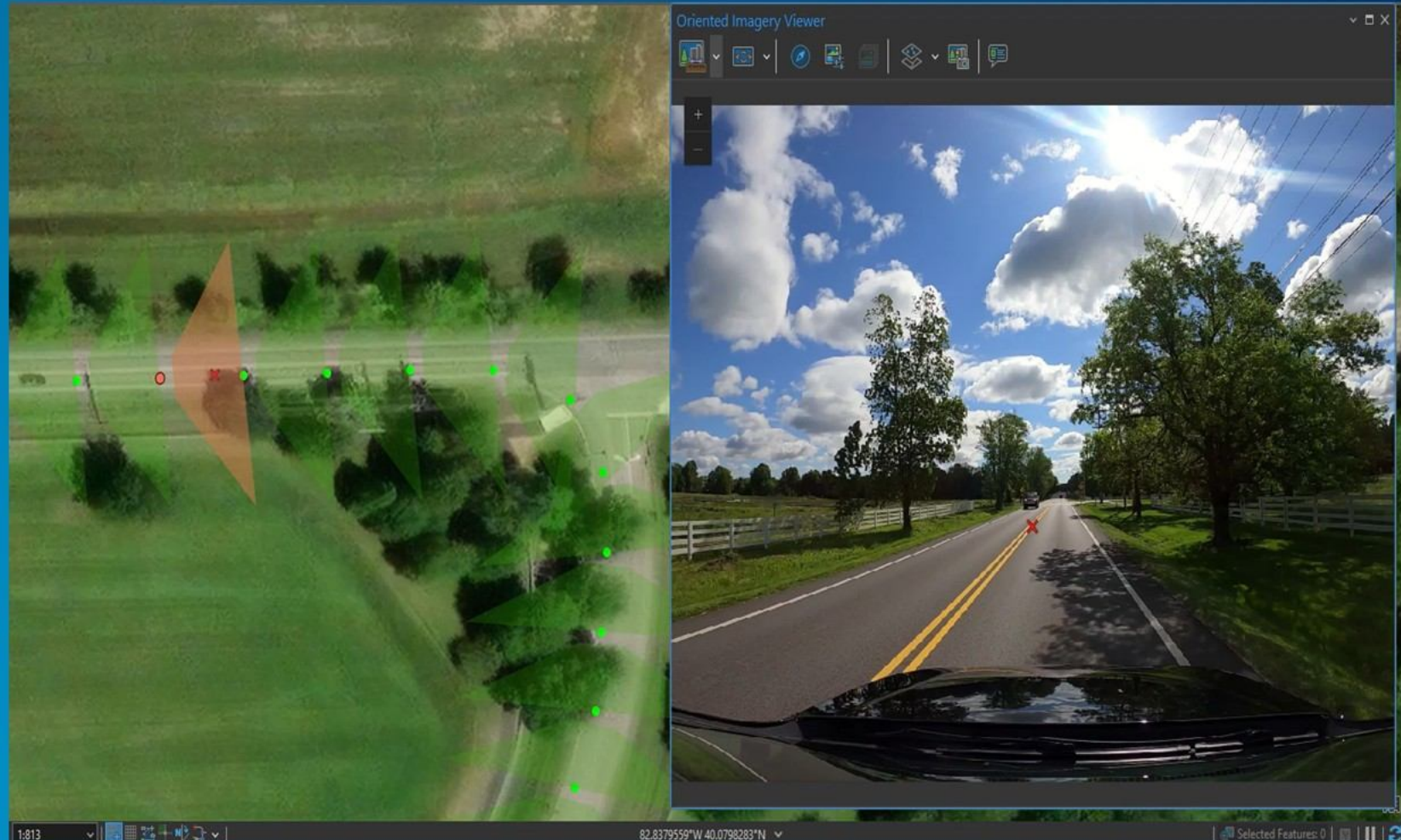
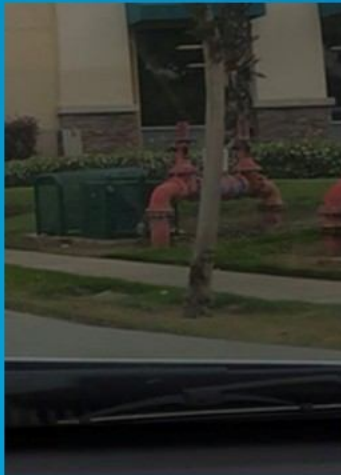
Integrated Ortho & LOD2 Buildings

A black hatchback car, possibly a Vauxhall Corsa, is parked on a paved street. A large, professional-grade camera rig is mounted on the roof of the car. The rig consists of a black camera body mounted on a complex metal frame with various adjustment points and a large black protective cover. The car's license plate is '880REY 75'. In the background, there is a building with a sign that reads 'Dunowell' in a large, serif font, with 'First Class Courthouse' written in smaller text below it. The building has large glass windows and doors. To the right of the car, there is a white metal gate and a small red fire alarm pull station on the wall.



Value of Street Level Oriented Imagery

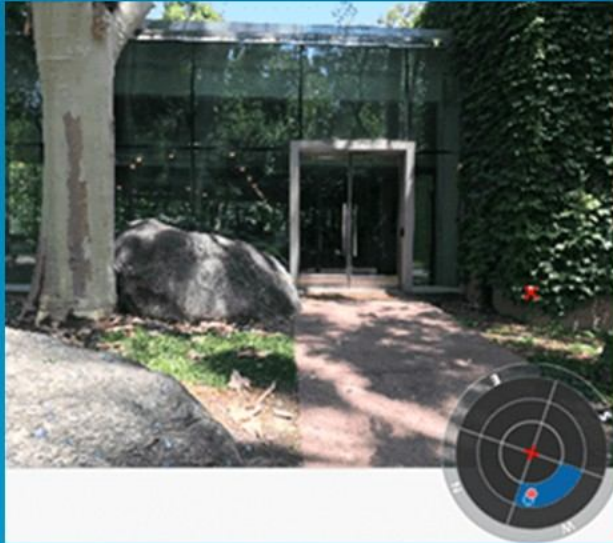
Turning pixels into information for spatial analysis



AI on Street-Level Imagery



Oriented Imagery



ArcGIS Pro - MyProject2 - Map

Feature Layer | Oriented Imagery

Project | Map | Insert | Analysis | View | Edit | Imagery | Share | Appearance | Labeling | Data | Data

admin (ArcGIS Enterprise) | ? | - | □ | ×

Clipboard | Navigate | Layer | Selection | Inquiry | Labeling | Offline

Contents

Search

Drawing Order

- Map
 - ☒ Esri Redlands Campus
 - ☒ World Boundaries and Places
 - ☒ Cobb County (Cyclomedia)
 - ☒ Exposure Points
 - ☒ Current Coverage
 - ☐ Additional Coverage
 - ☒ World Imagery

Imagery Viewer - Cobb County (Cyclomedia)

1:756 | 84.4752714°W 33.8837866°N | Selected Features: 0

Text-Prompted Discovery

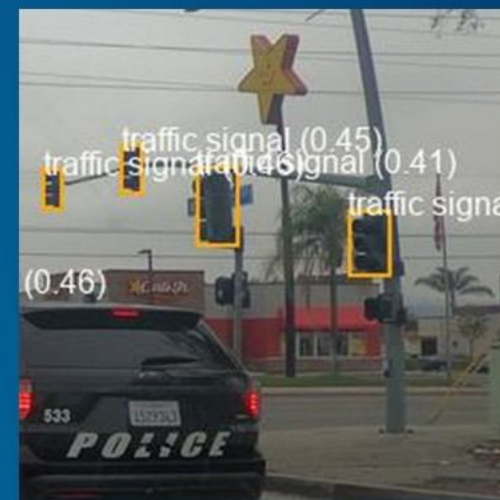
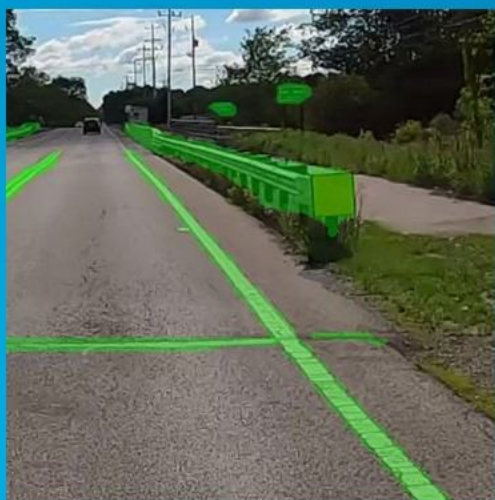


```
# Prompts - adjust to your targets
TEXT_LABELS = [
    # Crosswalks and stop control
    "zebra crosswalk, ladder crosswalk, continental crosswalk, marked pedestrian crossing, NOT lane lines, NOT centerlines",
    "stop bar, stop line at intersection, broad transverse bar, NOT lane lines, NOT dashed lines",

    # Symbols and word legends
    "turn arrow marking (left, right, straight, merge), chevron arrow on pavement, NOT lane lines",
    "bike lane symbol (bicycle glyph) on pavement, NOT lane lines",
    "HOV diamond symbol on pavement, NOT lane lines",
    "railroad crossing pavement marking 'RXR' with transverse bars, NOT lane lines",
    "word markings such as 'ONLY', 'BUS', 'SLOW', 'SCHOOL' painted on pavement, NOT lane lines",
    "speed hump markings (triangles/shark teeth), yield triangles on pavement, NOT lane lines",
    "handicap/ADA parking symbol on pavement, NOT lane lines",

    # Intersections & guidance markings (non-longitudinal)
    "dashed box intersection markings, keep clear box, painted curb extensions, NOT centerlines, NOT edgelines",

    # Parking and non-travel-lane
    "parking stall markings (T or L shapes), angled stall markings, NOT travel-lane lines"
]
```



Road Features

- Road signs
- Guard rails
- Curbs
- Road cracks
- Pavement markings
- Other road features

Dataset Disclaimer:

Although the Arizona Department of Transportation's data in AZGeo has been produced and processed from sources believed to be reliable, no warranty expressed or implied is made regarding accuracy, adequacy, completeness, legality, reliability or usefulness of any information. This disclaimer applies to both isolated and aggregate uses of the information. The Arizona Department of Transportation provides this information on an "AS IS" basis. All warranties of any kind, express or implied, including but not limited to the implied warranties of merchantability, fitness for a particular purpose, freedom from contamination by computer viruses and non-infringement of proprietary rights are disclaimed. Changes may be periodically added to the information herein; these changes may or may not be incorporated in any new version of the publication. If the user has obtained information from the AZGeo webpage from a source other than the AZGeo Home Page, the user must be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out-of-date. It is recommended that the user pay careful attention to the contents of any metadata associated with a file, and that the originator (MPD GIS) of the data or information be contacted with any questions regarding appropriate use. If the user finds any errors or omissions, we encourage the user to report them to the Arizona Department of Transportation GIS team at MPDGIS@azdot.gov.

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Source: ADOT
For demonstration purposes only

Lat / Long Inference and Detection Output from Single Images



Detection Results | Visual Results

I-91 0091 NB 97.700 BRADFORD 0091-0000 97.700 07 NYS 08/10/2013



R3-2



R1-2

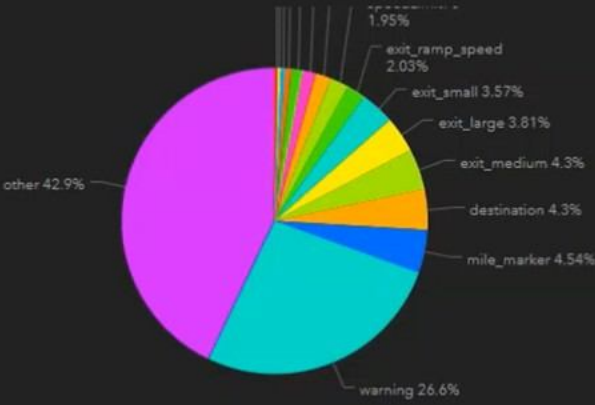
Deep learning classification of sign assets

Road Sign Classification Demo

For IN 015 NB

Detections
1,233

Sign Categories



road_works_ahead	4	speedLimit65	4	lane_exit	6	parking	8
stop_ahead	15	overhead_sign	17	route_marker	20	speedLimit75	24
exit_ramp_speed	25	exit_small	44	exit_large	47	exit_medium	53
destination	53	mile_marker	56	warning	328	other	529

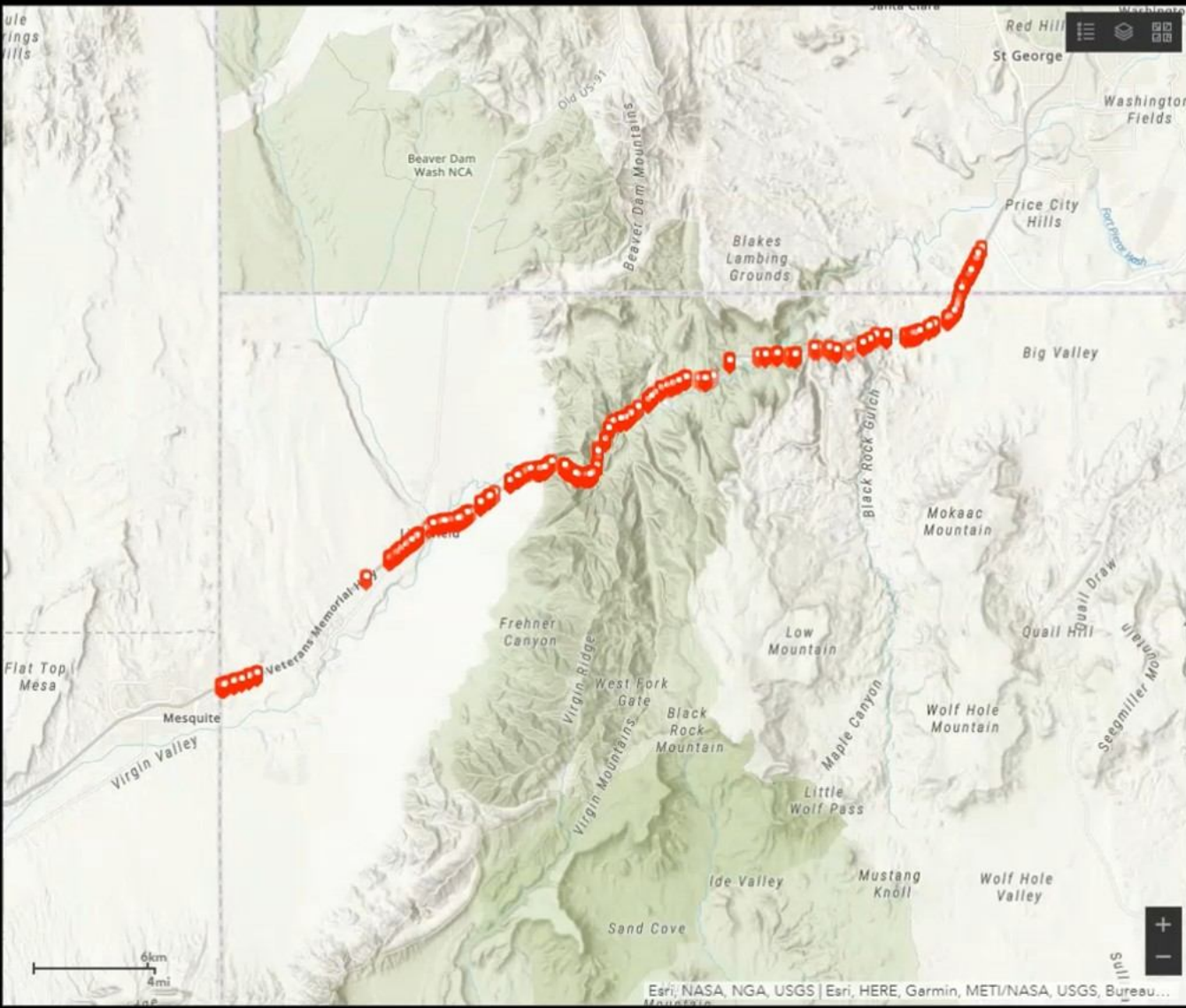
Last update: a minute ago

Last update: a minute ago

Detection Confidence



Last update: a minute ago



Road Features

- Road signs
- Guard rails
- Curbs
- **Road cracks**
- Pavement markings
- Other road features

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Training a Deep Learning model to detect different types of Road Cracks

Table 1: Road damage types in our dataset and their definitions.

Damage Type			Detail	Class Name
Crack	Linear Crack	Longitudinal	Wheel mark part	D00
			Construction joint part	D01
		Lateral	Equal interval	D10
			Construction joint part	D11
	Alligator Crack		Partial pavement, overall pavement	D20
Other Corruption			Rutting, bump, pothole, separation	D40
			Cross walk blur	D43
			White line blur	D44

Source: Road Maintenance and Repair Guidebook 2013 JRA (2013) in Japan.

Note: In reality, rutting, bump, pothole, and separation are different types of road damage, but it was difficult to distinguish these four types using images. Therefore, they were classified as one class, viz., D40.

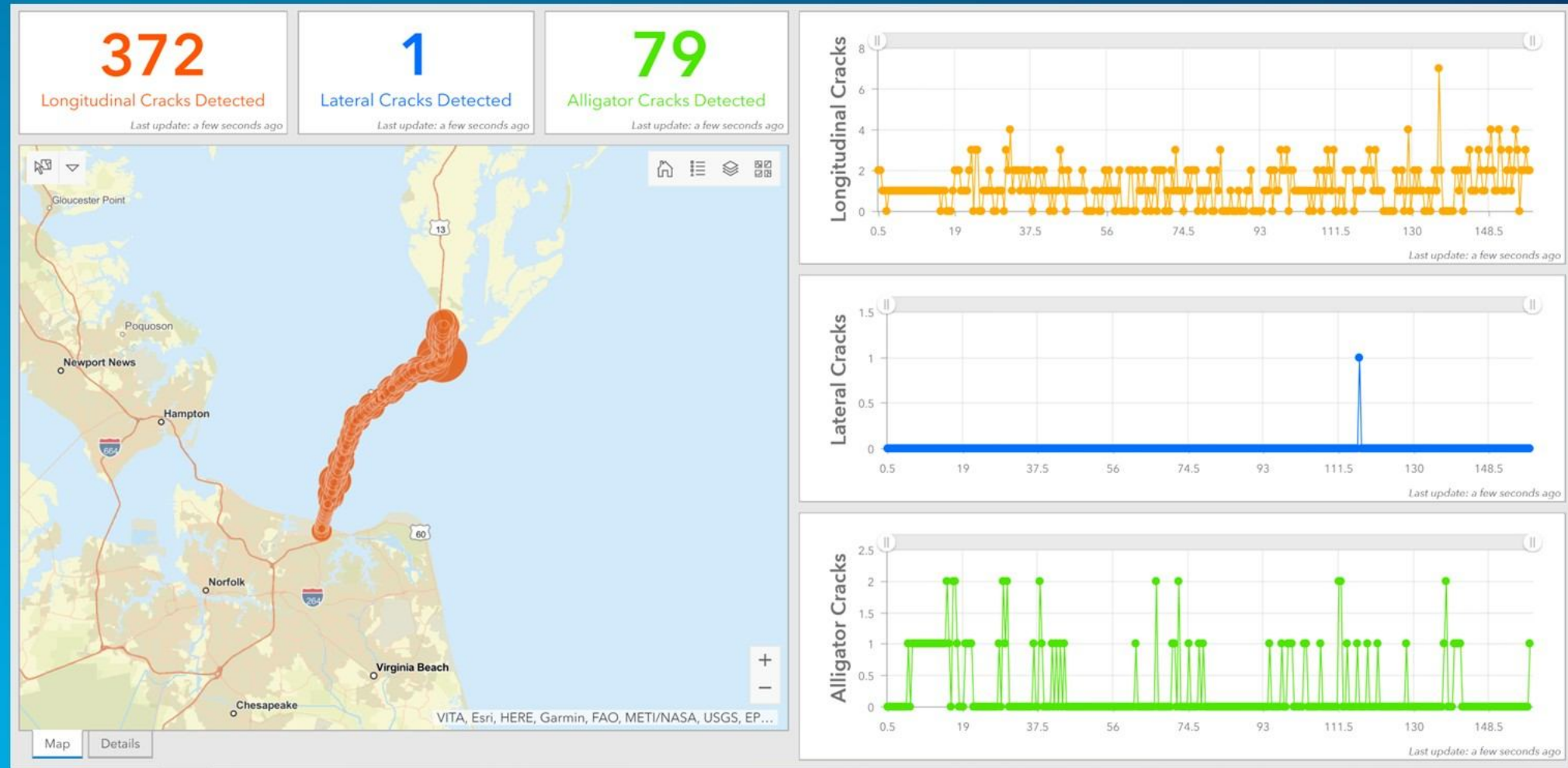


Real-Time Detection of Cracks (Video)



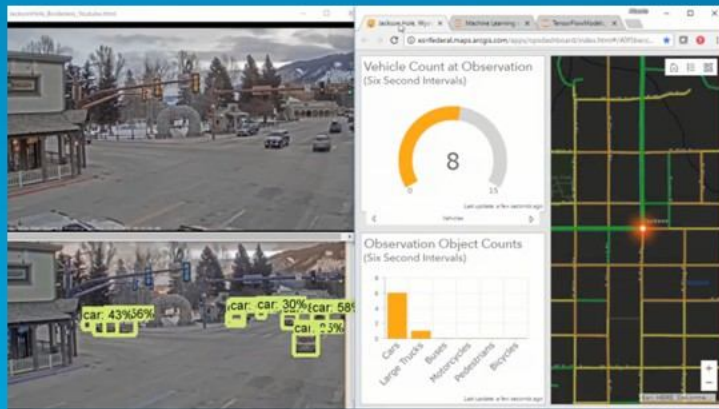
Source: Esri Industries
GeoAI for Transportation Webinar (2021)

Visualizing Areas with Highest amount of Cracks



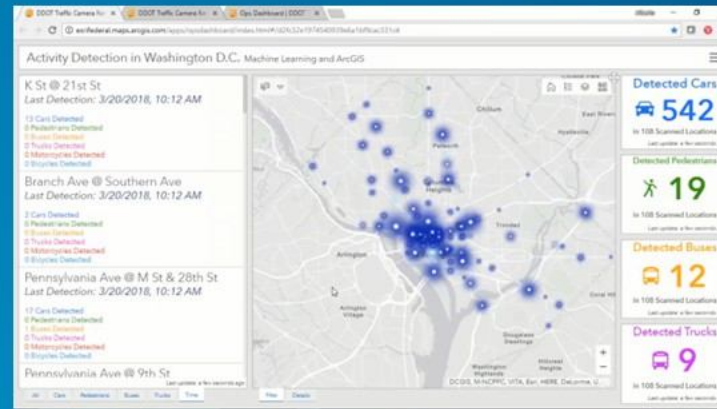
Introduction-Video AI

Transportation Authorities can extract a lot from Street Cameras by using AI. Real-Time Traffic Patterns, Congestion Analysis, Adaptive Signal Control, Incident Analysis, and Anomaly Detection are just examples. This could help **Optimize Traffic**, **Better Planning for Infrastructure**, and **Proactive Handling of Incidents**



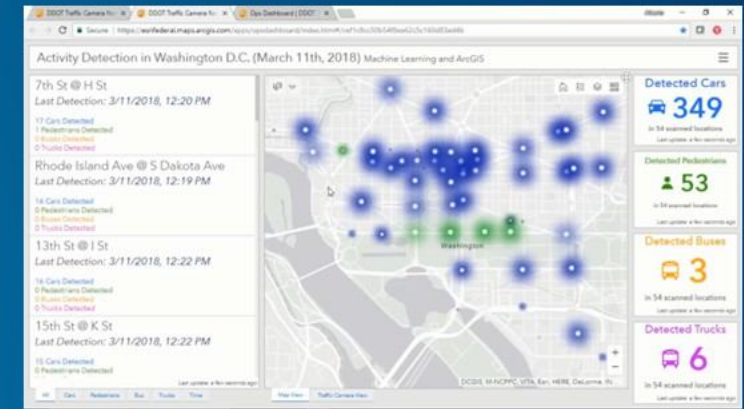
Real-Time Object Detection (by Type)

Detect Cars, Trucks, Buses, Motorcycles, Bicycles, Pedestrians, and more. Analyze Geospatial Behavior in Ops Dashboard



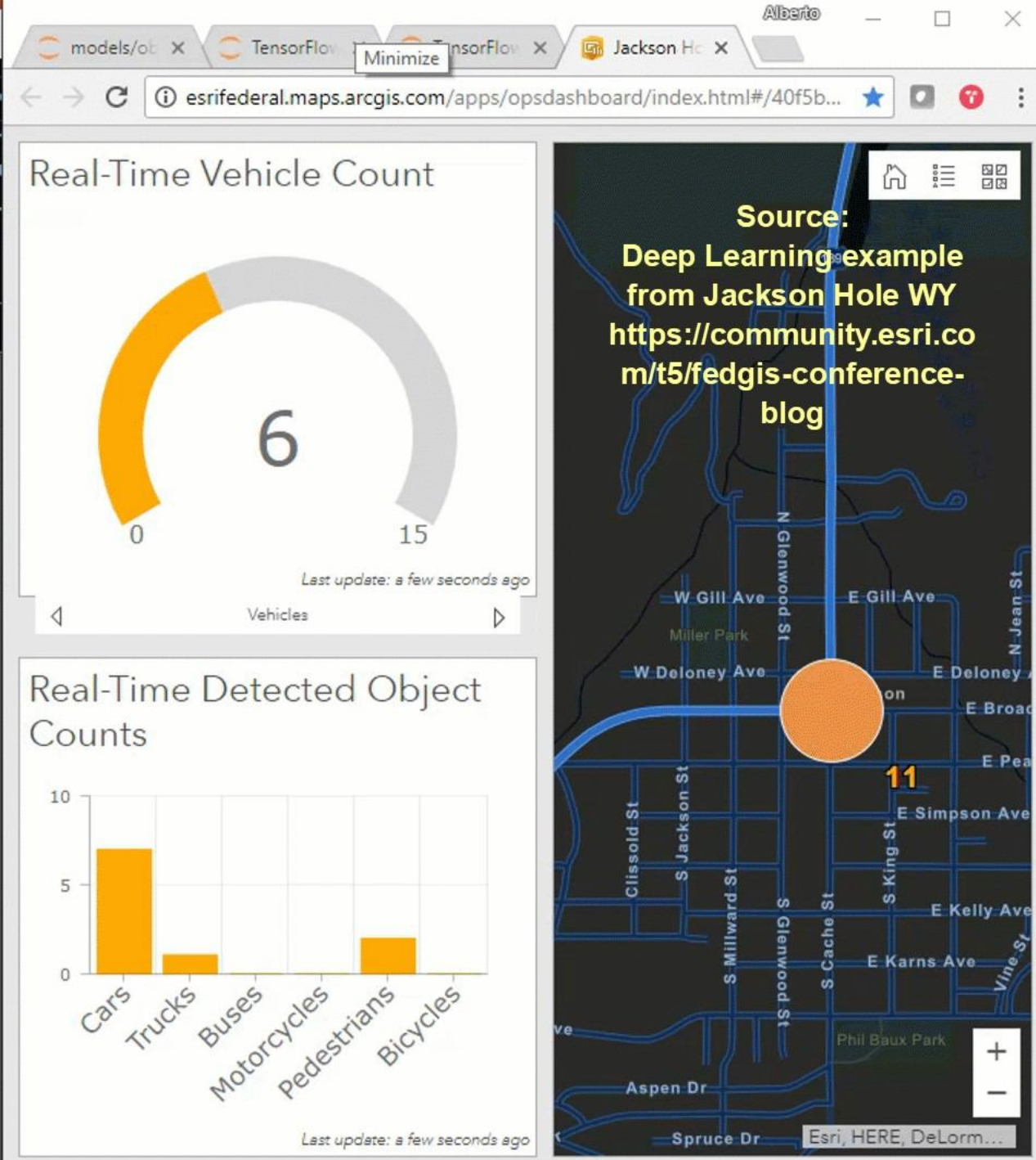
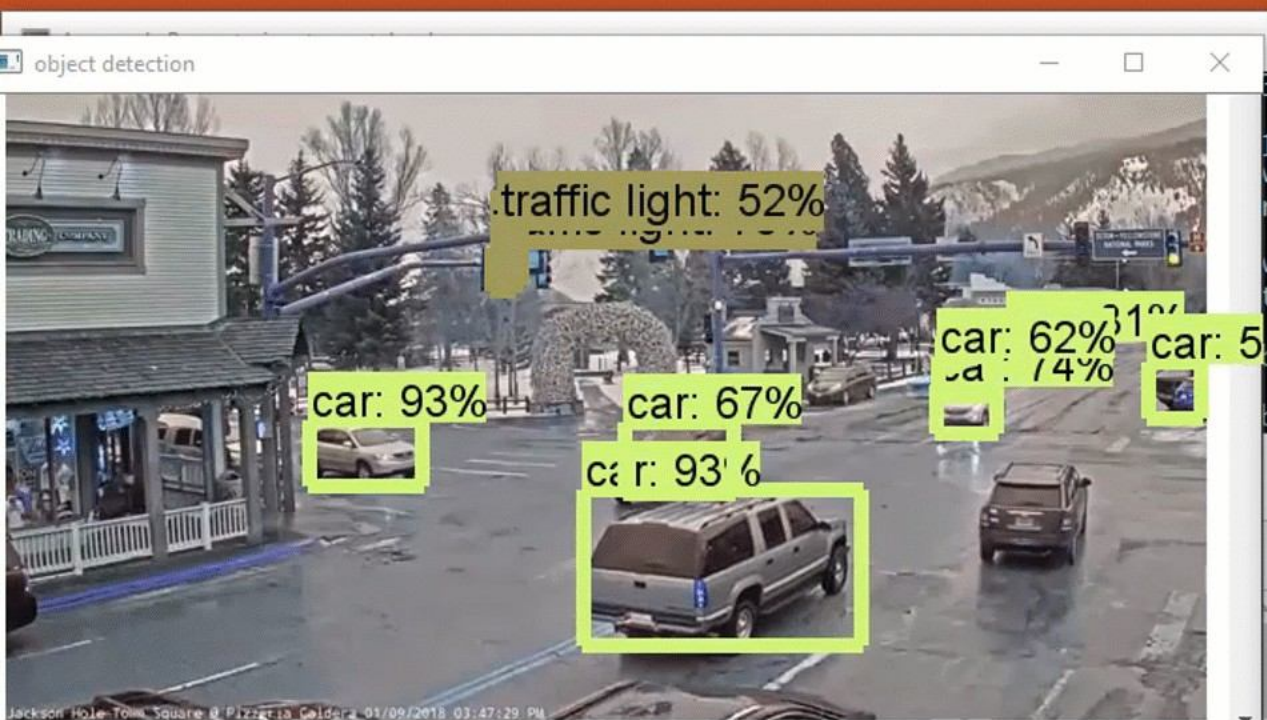
City Wide Traffic Analysis

Analyze the movement per Vehicle Type per area in Real-Time. Zoom in to any Intersection/Area to dig deeper into what's happening



Anomaly Detection

Detect Abnormal Behaviors & Anomalies by contrasting Real-Time Traffic Patterns with Historical Ones





AI for Disaster Response

Detect Damaged Buildings, Detect Damaged Roads, & Allocate Resources

Source:

<https://www.esri.com/arcgis-blog/products/arcgis-enterprise/public-safety/a-i-enabled-disaster-response>



Detect damaged structures

An A.I model runs on aerial imagery and outputs a feature class of damaged and undamaged structures



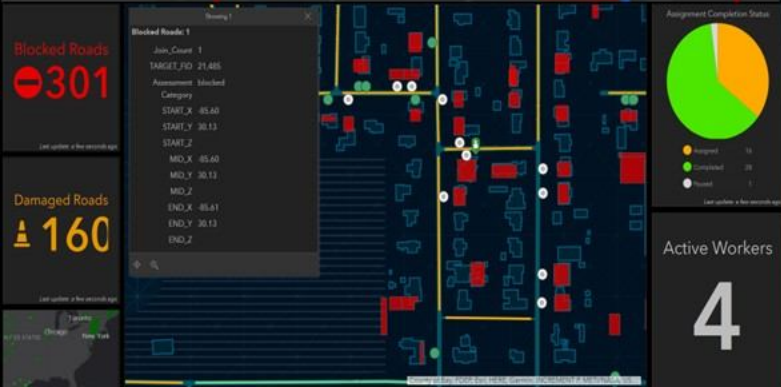
Detect damaged roads

An A.I model runs on aerial imagery + road layer and outputs a feature class of damaged and undamaged roads



Optimal routing

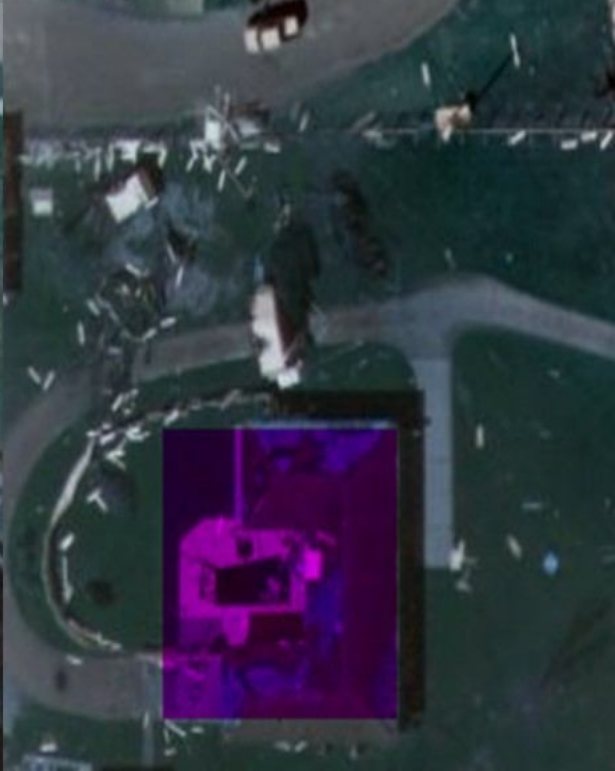
ArcGIS Network Analyst consumes the damaged structures and blocked roads feature classes, and creates an optimal route from a base (fire station, shelter etc.) to all the damaged structures, accounting for blocked roadways



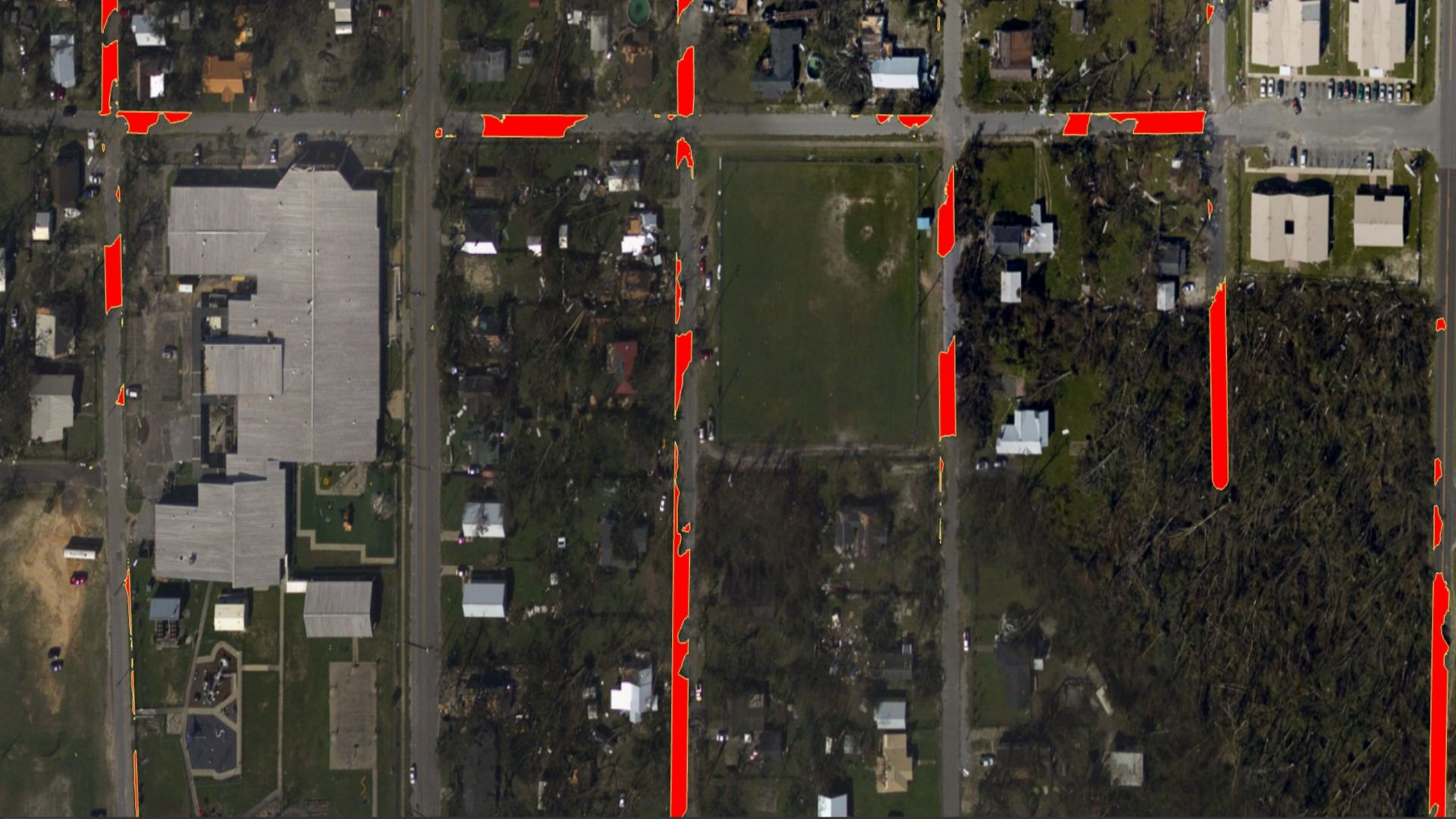
Operational dashboard

Utilizing the Esri Workforce mobile app, we can assign each responder a set of structures to inspect and use Operations Dashboard to monitor their progress and other key metrics









Blocked Roads

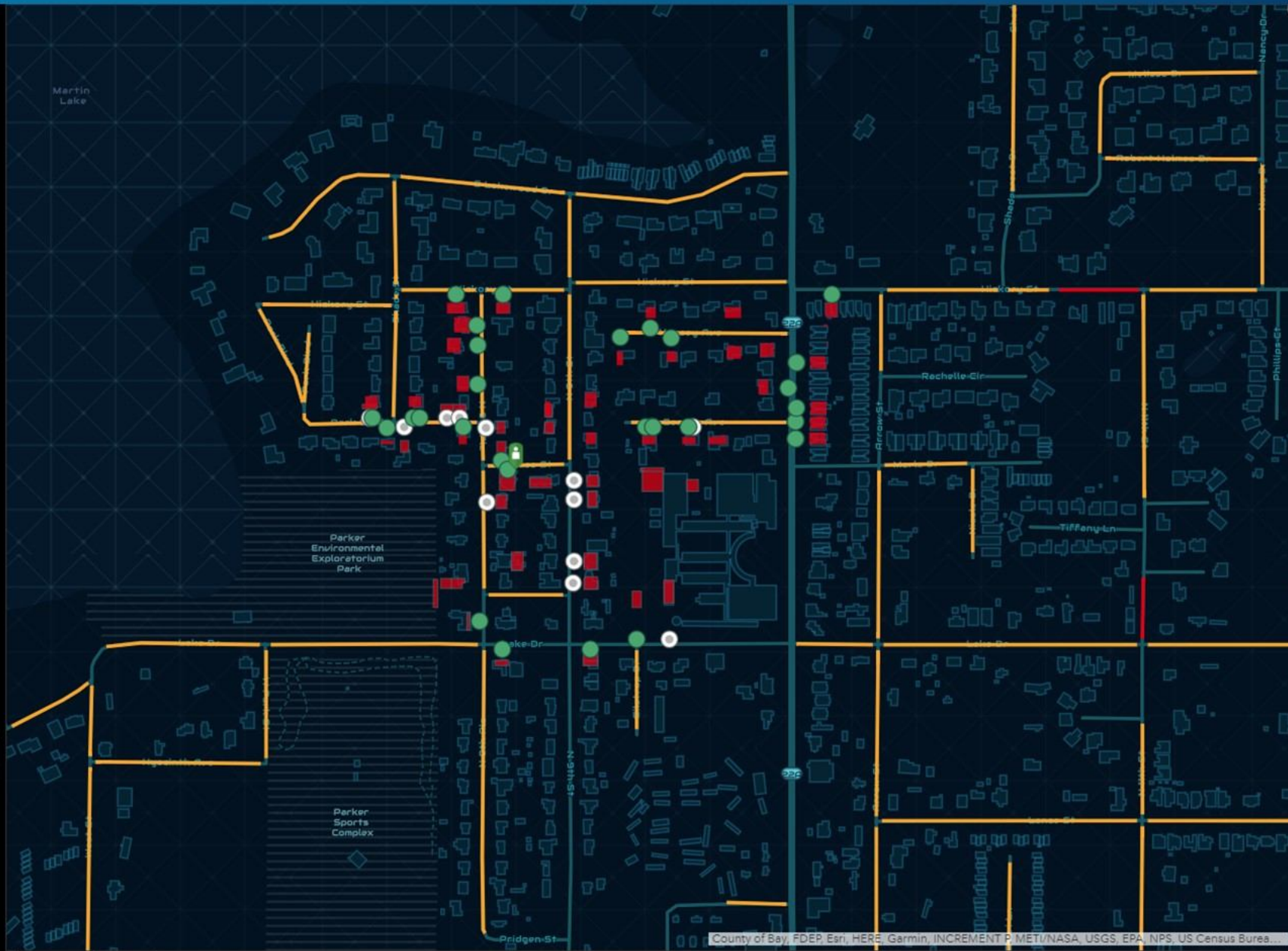
301

Last update: a few seconds ago

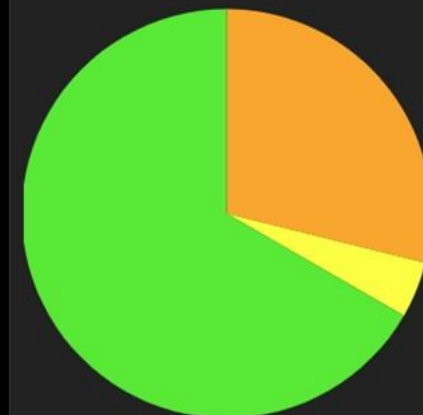
Damaged Roads

160

Last update: a few seconds ago



Assignment Completion Status



Status	Count
Assigned	13
In Progress	2
Completed	30

Last update: a few seconds ago

Active Workers

3

County of Bay, FDEP, Esri, HERE, Garmin, INCREMENT P, METI/NASA, USGS, EPA, NPS, US Census Bureau...

Rethinking AI for Disaster Support

Use Generative AI with post-disaster imagery

Let the model describe the image

This is an aerial image of a site impacted by hurricane. I want you to look at the picture and let me know if you see any debris in this.
Answer with a Yes or No and also share the reason why you think so. If the image is not clear at all, choose No as the response.

Geoprocessing

Interrogate Imagery with GeoPrompts

Parameters Environments

Input Raster
NOAA_Helene.crf

Features to Interrogate
FLParcels

Query Prompt
This is an aerial image of a site impacted by hurricane. I want you to look at the picture and let me know if you see any debris in this.
Answer with a Yes or No and also share the reason why you think so. If the image is not clear at all, choose No as the response.

Max Tokens 30

Model
GPT-4o



Yes. The image shows what appears to be a structure with significant damage. There is scattered material around the site, which suggests that debris is present

No, I do not see any debris in this image. The area appears to be clear and the structures and surrounding grounds look intact.

ArcGIS Is Being Enhanced with AI

Integrating the power of GIS and AI across organizations and workflows



AI Tools & Models

*Advancing the
Science of GIS*

Science



AI Assistants

*Enhancing user
experiences*

Productivity



Agentic AI

*Automating tasks &
expanding GIS*

Impact



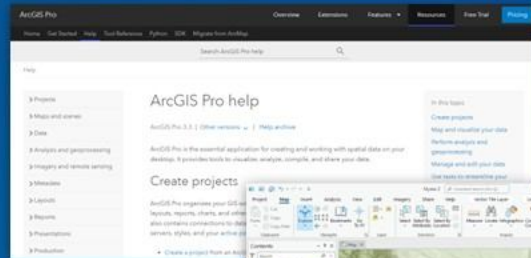
ArcGIS

- GIS Tools & Data
- Agents
- Models
- LLMs

ArcGIS AI Assistants

Leveraging Generative AI to streamline your work and make GIS easier for everyone

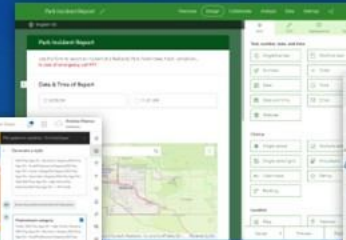
Documentation



Mapping



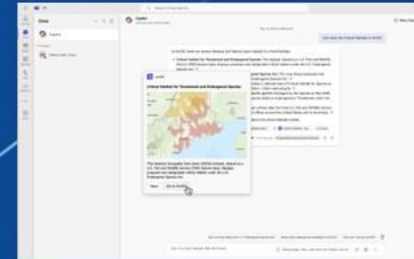
Surveys and Data Collection



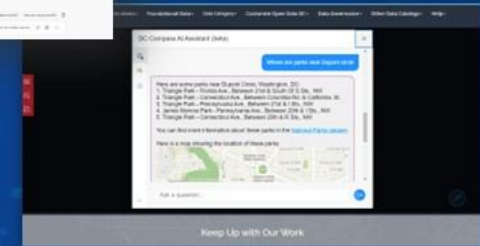
Software Developers



Collaboration



Citizen Query and Engagement



Analytics

Boosting productivity
and making GIS easier

ArcGIS



AI Assistants

Streamlining your work and making ArcGIS easier

Arcade (Code Generation)



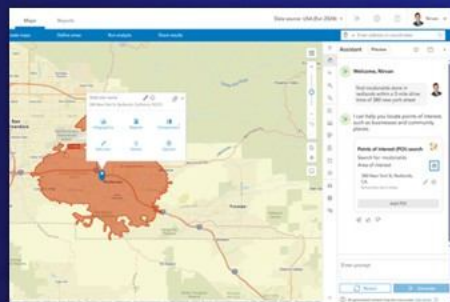
Solutions (Discover and Deploy)



Documentation (All Products)



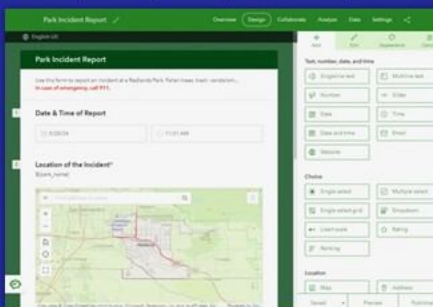
Business Analyst (Analysis & Mapping)



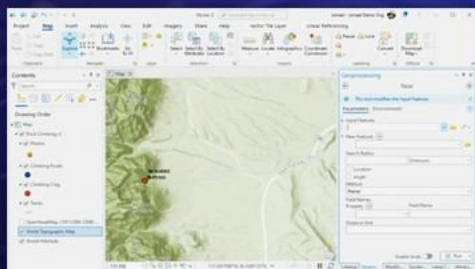
ArcGIS (Copilot Integration)



Survey123 (Survey Creation)



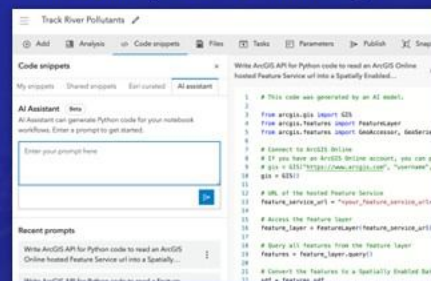
Pro (User Productivity)



StoryMaps (Storytelling, Writing)



Notebooks (Code Generation)



Apps

- Pro
- StoryMaps
- Hub
- Business Analyst
- Instant Apps
- Survey123
- Field Maps
- Drone2Map
- Microsoft Teams
- Notebooks
- ArcGIS Solutions
- ...

*Powered by natural language
and generative AI*

... Helping users be more productive

ArcGIS Is Being Enhanced with AI

Integrating the power of GIS and AI across organizations and workflows



AI Tools & Models

*Advancing the
Science of GIS*

Science



AI Assistants

*Enhancing user
experiences*

Productivity



Agentic AI

*Automating tasks &
expanding GIS*

Impact



ArcGIS

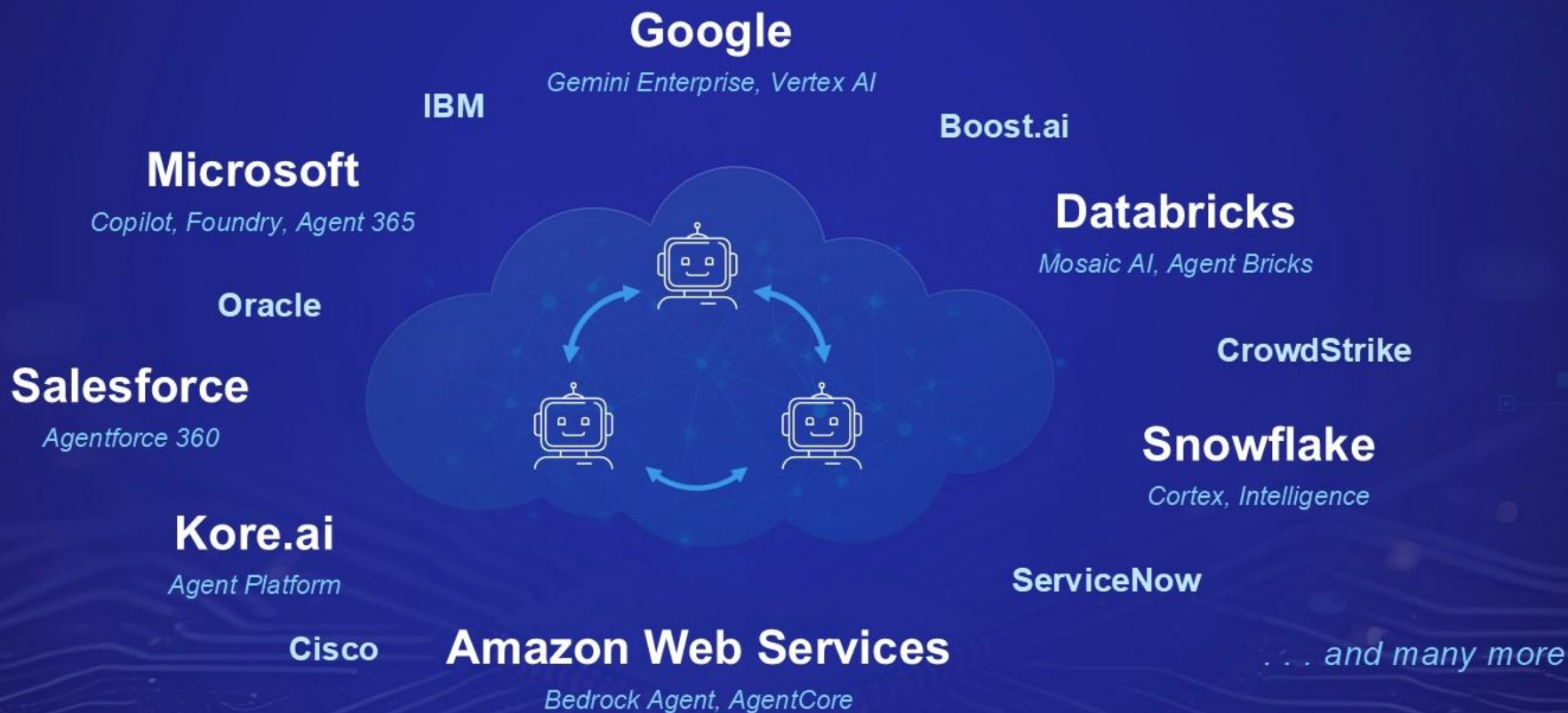
- GIS Tools & Data
- Agents
- Models
- LLMs

Reactive vs **Proactive**

Tasks vs **Workflows**

Agentic AI Is Emerging Rapidly

Many agentic AI platforms are being developed and adopted



Creating a new type of computing system

ArcGIS Is Becoming Agentic

Expanding the reach and impact of GIS

Two key protocol types emerge to facilitate interoperability.

One connects agents to tools and resources.

The other enables agent-to-agent collaboration.

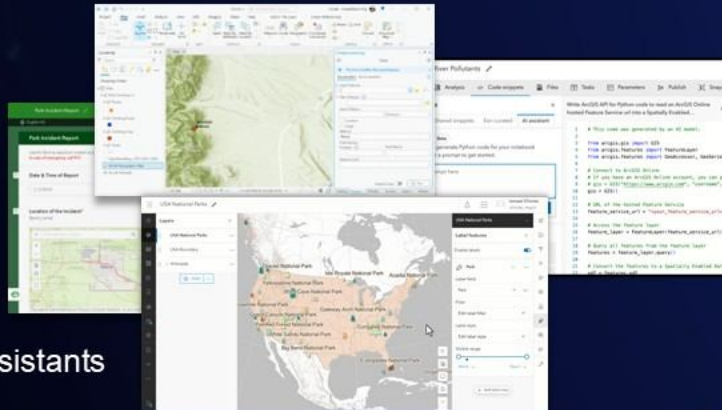
The Agent2Agent (A2A) Protocol and the

Model Context Protocol (MCP)

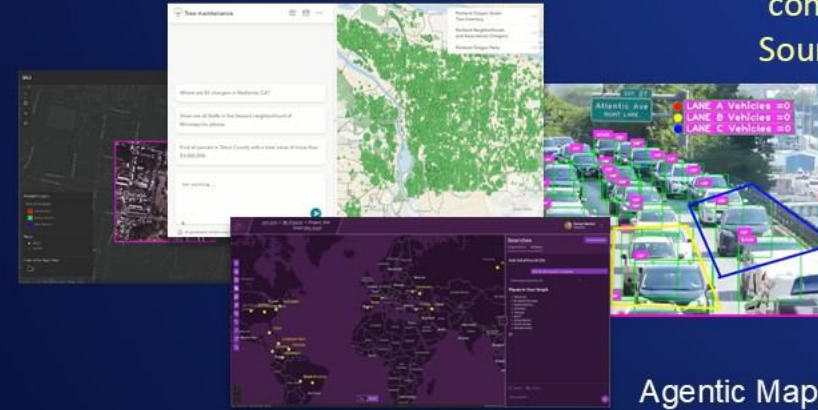
address these distinct but highly complementary needs.

[Source-a2a-protocol.org](https://source-a2a-protocol.org)

AI Assistants



Agentic Mapping Apps



ArcGIS



MCP

Geospatial
Capabilities

A2A



Other Agentic Platforms

ArcGIS for Agentic AI

Supporting geo-enabled and geo-centric workflows



GEO-ENABLED

"Bringing GIS to agentic ecosystems"

GEO-CENTRIC

"Bringing agentic AI to GIS workflows"

Model Context Protocol (MCP)

An interoperability standard for AI apps to work with external systems

ArcGIS



MCP



Agent Orchestration

What Is a Geospatial AI Agent?

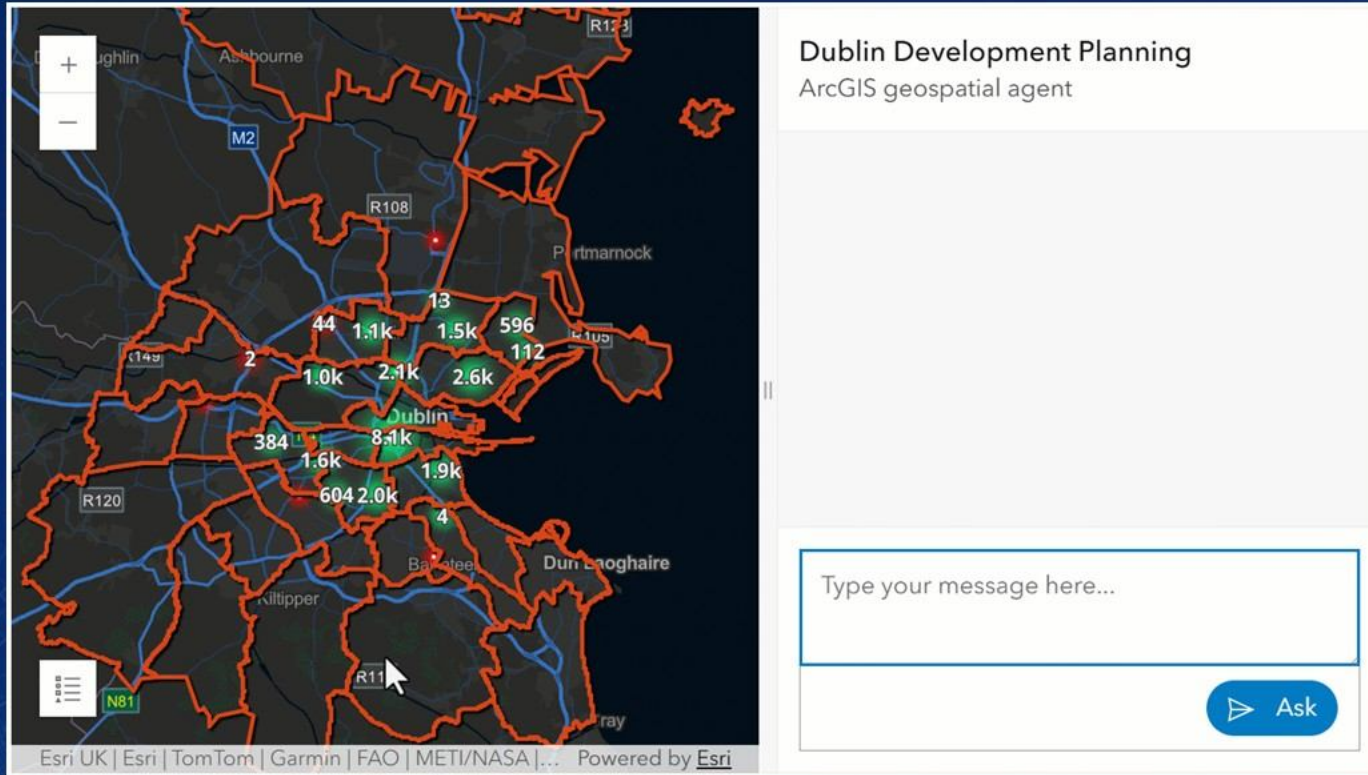
A **geospatial AI agent** is a type of agent that has been specifically configured to handle geospatial tasks.

These agents understand the foundational principles of geography, GIS tools and geospatial data.



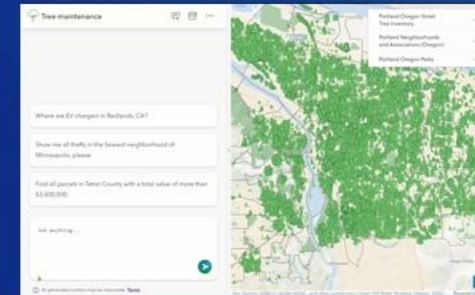
Building Agentic Mapping Apps

Streamline workflows and simplify access to geospatial knowledge



Natural language UX powered by geospatial agents

Configure apps (No Code)



Data Explorer
Instant App
Experience Builder

Code custom apps

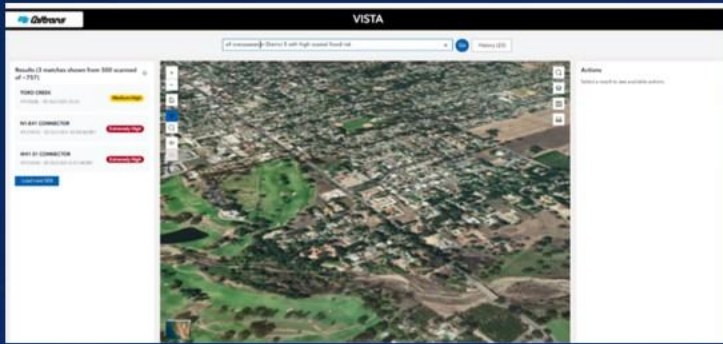


ArcGIS Maps SDK
for JavaScript

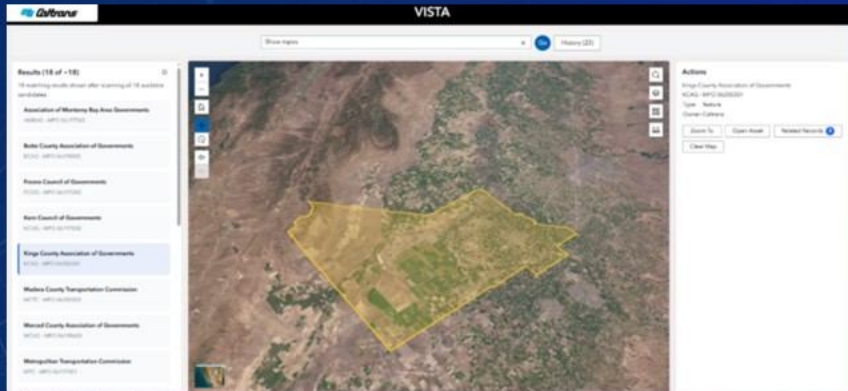
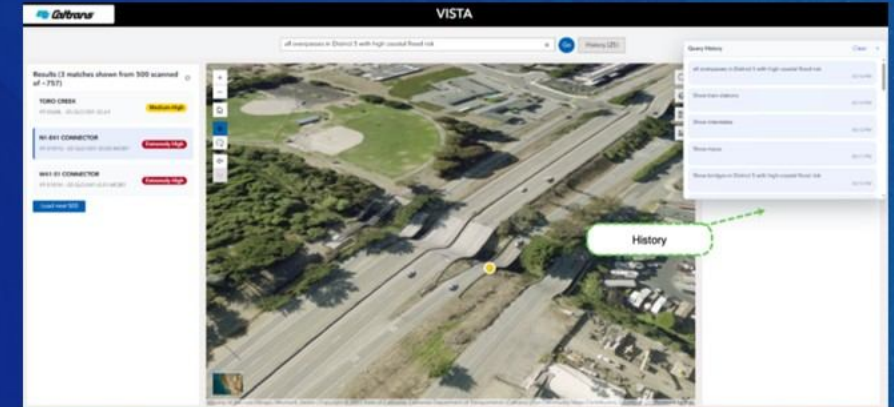
Expanding the reach of GIS

Agentic AI prototype from Caltrans

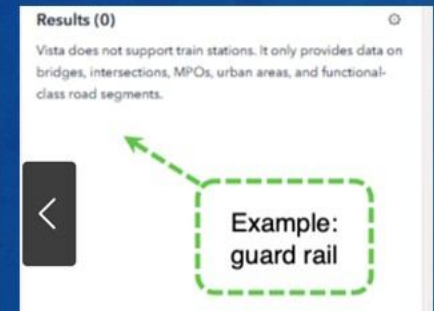
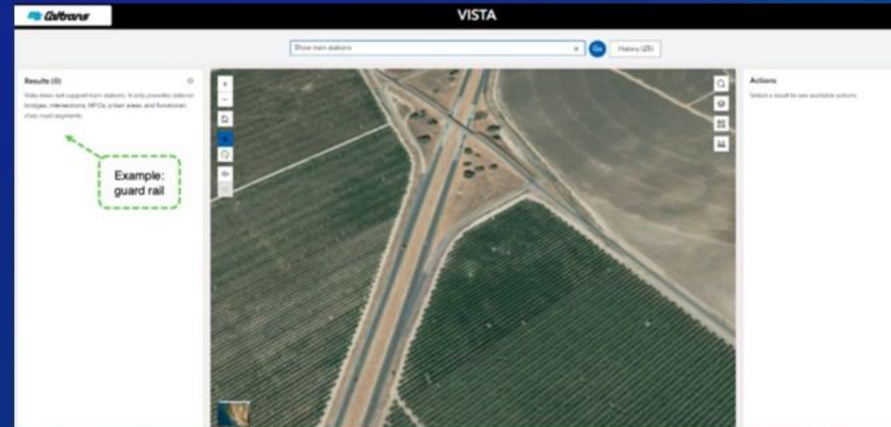
VISTA app for HPMS AI prototype



Show overpasses
In District 5
with high risk of
coastal flooding



Show
MPO's



Show train stations-zero results

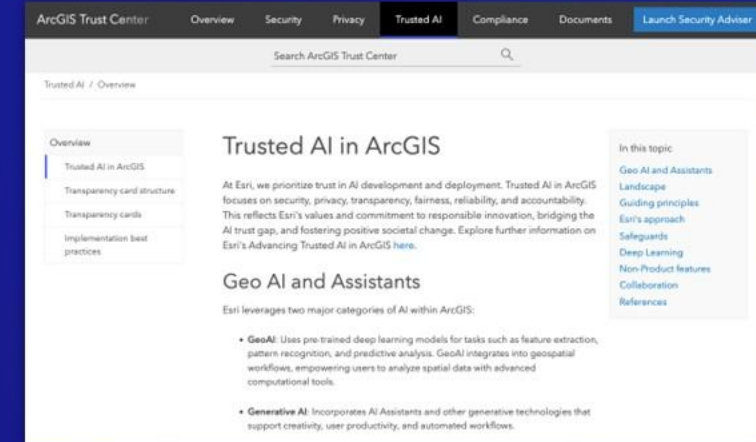
Trust . . .

Trusted AI in ArcGIS

Driven by AI principles and standards

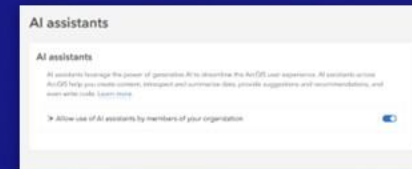
Trust center website

- Explainable, fair and reliable
- Model / algorithm transparency
- Secure and aligned with policies
- Preserve data security, use terms



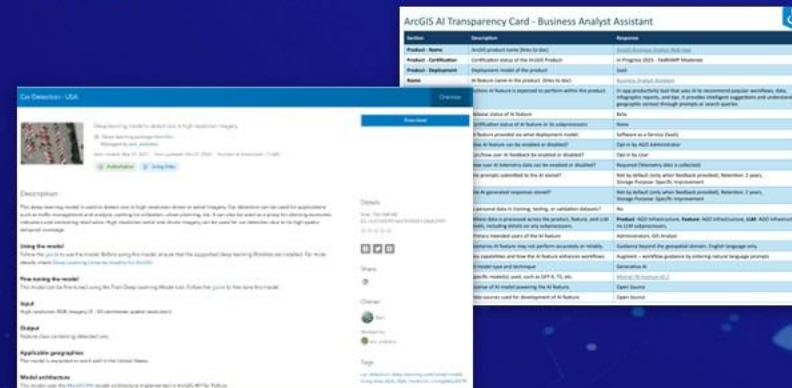
AI is a shared responsibility

- Settings, monitoring, training, validation

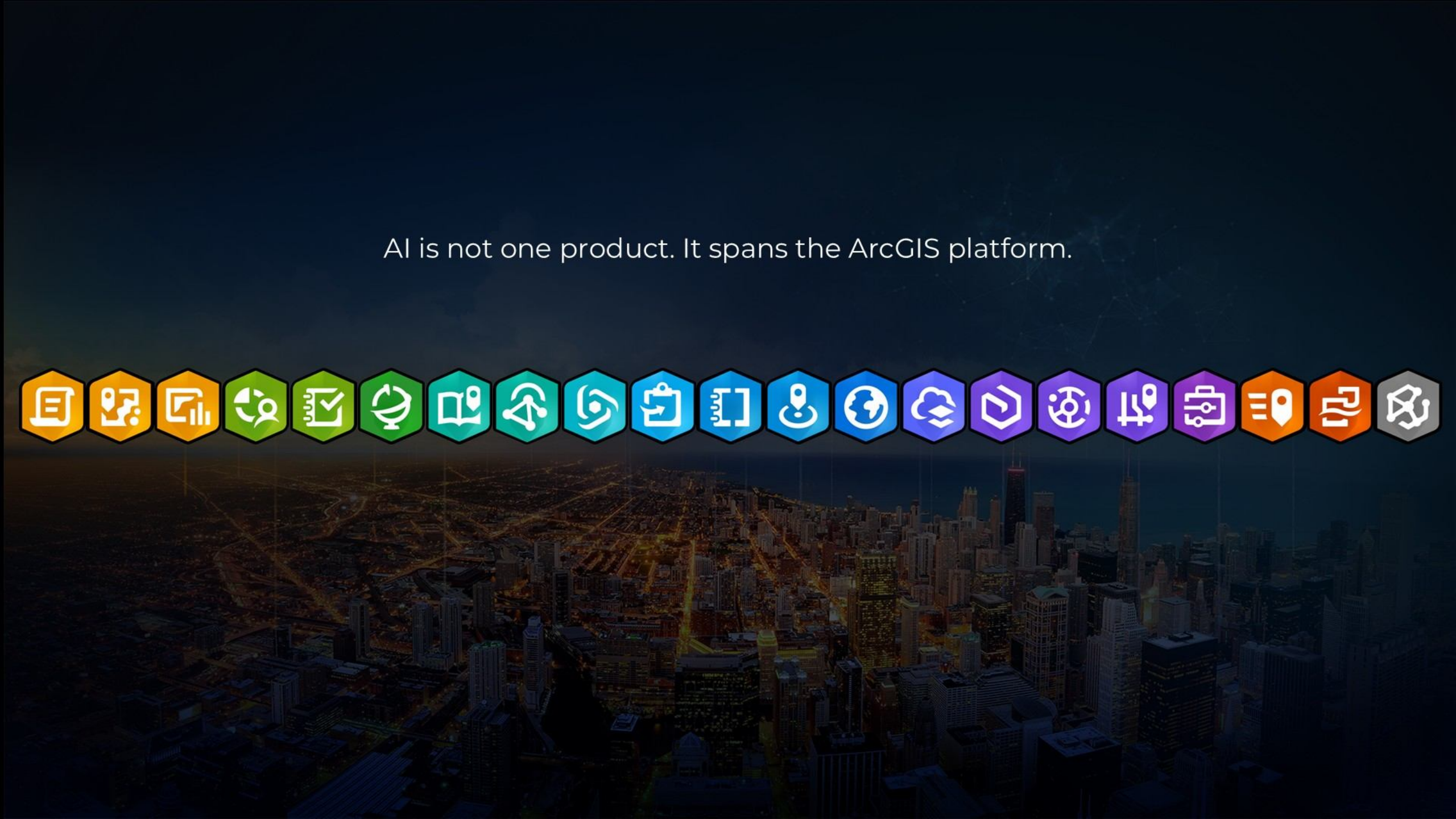


AI metadata

- Model cards
- Transparency cards



AI is not one product. It spans the ArcGIS platform.

A horizontal row of 20 hexagonal icons, each containing a white symbol representing a different ArcGIS AI or analytics tool. The icons are arranged in a single row and are color-coded: orange, green, blue, and purple. The background of the slide is a dark, high-angle view of a city at night, with glowing yellow lines representing data or network connections overlaid on the cityscape.

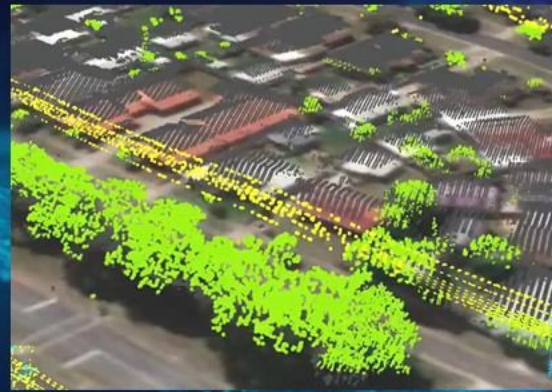
What Value Does AI Bring to ArcGIS?

GENERATE EFFICIENCIES



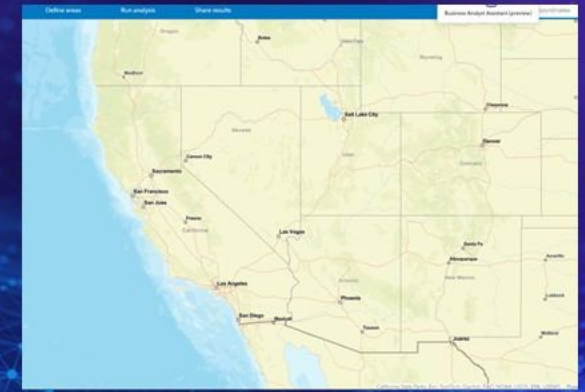
Connect Business Systems
Democratize Data Creation
Automate Manual Tasks

ACCELERATE ANALYSIS



Deeper Insights
Visualization tools
Predictive Spatial Analytics

ENHANCE USER EXPERIENCE



Intelligent Experiences
AI-enhanced mobile and office tools
Virtual and In-App Assistants



Thank you!



THE SCIENCE OF WHERE®